Thinned Nonhomogenous Poisson Processes for Presence-Only data

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Welcome

This is a discussion about using Nonhomogenous Poisson Point Processes for modelling Presence-Only data. It reviews the Poisson Point Process, Presence-Only data, data generation using the process, and relevant modelling details using the Bayesian programming language Stan.

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1 Introduction

In this document we discuss models for presence-only data, using Thinned Nonhomogenous Poisson Processes and Log Gaussian Cox Processes. The initial planned layout is as follows: Introduce Poisson Process models and the data generating process. Then, proceed to fit a Thinned Nonhomogenous Poisson Process in Stan (and in spatstat), on simulated presence-only data.

The rest of this document implements the above description, including theory where necessary.

2 Poisson process

The poisson process describes a set of points S where $\{s_i\} \subseteq D$ with intensity λ. Given an area or quadrat called subregion A of D of size $|A|$ centered around site s_i we can model the number of individuals occurring at that site $N_s(A)$ with a poisson distribution. This distribution will have mean

$$\Lambda(A) = \int \lambda(s)ds$$

For the demonstration here, we assume $|A| = 1$. However, the parameters in the Poisson Point Process are invariant to the scale of the data.

Typically, we model the intensity with a log-linear function, for example with parameters α and β,

$$\lambda(s) = \exp[\alpha + \beta' x(s)]$$

so that the expected number of observations in region A is now

$$\Lambda(A) = \int_A \exp[\alpha + \beta' x(s)]ds$$

We quickly notice that this integral is intractable and we will need a method of numerical integration. Indeed this will be discussed and explored here.

Consider, however, that we are going to be dealing with environmental data with environmental covariates, which are going to be discretized at some resolution inherently. Therefore, we can think of the inhomogenous poisson process as a continuous limit of a discretized set of conditionally independent poisson random variables, where the IPP emerges as the discretization grows smaller, i.e., $\Lambda(ds), ds \to 0$. In fact, using this continuous limit is how the likelihood for the PP can be derived (Banerjee, 2014. p203-4). However, considering that we will be working in a discretized space, the continuous framework ends up being less practically useful.

We now define the likelihood for the NHPP (Banerjee, 2014, p214)

$$f(s_1, s_2 ... s_n | N(D) = n) = \prod_i \frac{\lambda(s_i)}{(\lambda(D))^n}$$

and the joint density will be

$$f(s_1, s_2 ... s_n, N(D) = n) = \prod_i \frac{\lambda(s_i)}{(\lambda(D))^n} \left[(\lambda(D))^n \frac{\exp(-\lambda(D))}{n!} \right]$$

where we can see the second term on the right corresponds to the poisson likelihood for the number of total observations in space D. The likelihood will then be

$$L(\lambda(s)|s_1, s_2, ..., s_n) = \prod_i \lambda(s_i) \frac{\exp(-\lambda(D))}{n!}$$

We can move between the continuous and discrete versions of this likelihood by partitioning D into a grid of c cells and taking the poisson likelihood

$$L(\lambda) = \prod_c (\lambda(A_c))^{N(A_c)} \exp(-\lambda(A_c)) \tag{2.1}$$

Noticing that we can sum all the exponents, we get

$$L(\lambda) = \prod_c (\lambda(A_c))^{N(A_c)} \exp(-[\lambda(A_1) + \lambda(A_2) + \cdots + \lambda(A_c)) = \prod_c (\lambda(A_c))^{N(A_c)} \exp(-\lambda(D)]$$

and $N(A_c) = 1/0, |A_c| \to 0$. The term on the right indeed reduces to 2.1. We can then safely work with the Poisson likelihood given our grid is fine enough. This will be explored later.

We still have, however, one issue. We originally defined the mean of the of region A according to the properties of the PP,

$$\Lambda(A) = \int_A \lambda(s) ds = \int_A \exp[\alpha + \beta' x(s)] ds$$

Suddenly we are saying that we can just work with the intensity for region A without clarifying what we should do about the integral over the functional for intensity. Unfortunately, while we may want the integral over s, we don't have information at the resolution of ds. All we have are covariates $\beta' X$. That is,

$$\int_A \exp[\alpha + \beta' x(s)] ds \approx |A| \exp[\alpha + \beta' x]$$

Notice that we no longer have $x(s)$. We assume the covariate information is relative constant over region A. If this doesn't hold, our approximation will not be correct. But, given that it is all the information we have, the assumption is built into any modelling procedure and is less an issue with the Poisson Process and more just an issue of covariate resolution which is a common modelling issue in spatial-temporal statistics. As is stated in Banerjee, 2014 (p216) "In the absence of finer covariate resolution, we cannot do better with regard to the ecological fallacy."

Also, originally, we mentioned that it is convenient to assume $|A| = 1$. This is now the reason, so we don't have to worry about the scaling factor.

Given this approximation and scaling assumption, we now have

$$\Lambda(A) = \int_A \exp[\alpha + \beta' x(s)] ds \approx |A| \exp[\alpha + \beta' x] = \frac{|D|}{c} \exp[\alpha + \beta' x]$$

where c is the number of cells and $|D|$ is our total area. Then the counts in subregion A will be distributed as

$$N(A_i) \sim Poisson\left(\frac{|D|}{c}\exp[\alpha + \beta' x_i]\right) = Poisson\left(\exp[\alpha + \beta' x_i]\right)$$

if the number of cells is equal to the area ($|A| = 1$). Based on this, we end up with the log-likelihood for our entire presence-only dataset:

$$l(\alpha, \beta) = \sum_{i \in B} N(A_i)(\alpha + \beta' x_i) - \frac{|D|}{c}\sum_{i \in B}\exp[\alpha + \beta' x_i] - \sum_{i \in B}\log(N(A_i))!$$

where $i \in B$ refers to the set of background points $B = 1, 2, ..., c$. We take this likelihood over the entire set of background points in order to appropriately estimate the integral

$$\Lambda(D) = \int_D \lambda(s)ds$$

We need all points in the grid to approximate the integral, as is clear in the original likelihood. Thus, we have defined our model and the likelihood that we can use for PO data.

```
library(ggplot2)
library(viridis)
#library(hrbrthemes)
library(cmdstanr)
library(bayesplot)
library(spatstat)
library(dplyr)
library(tidyr)

source("presence_only_functions.R")
source("surface_functions.R")
eval_pp <- T
eval_lgcp <- F
simple_surface <- T
```

2.1 Presence-only data generation

Having introduced some theory for Poisson Point Process models, we next move on to data simulation. Here we simulate some presence-only data, in preparation to fit a nonhomogenous poisson process model. We specify the intensity for each site, the total number of individuals, the area of each site (assumed to be 1 here), and the sampling bias.

Following Fithian 2015, each observation i is associated with a site s_i area D. For each site s_i, there are associated covariates $x_i = x(s_i)$ and $z_i = z(s_i)$. For each survey site, s_i represents the centroid of a quadrat Ai. This is an essential assumption in our model, where we will use the Poisson likelihood to for the number of counts in each quadrat. At s_i we observe counts $N_i = N(A_i)$ or binary presence/absence indicators y_i, with $y_i = 1$ if $N_i > 0$ and $y_i = 0$ otherwise.

One issue in generating the data is that in using a poisson model to simulate the counts for each site, we lose individual point information in aggregating the points to counts per quadrat. However, this issue of aggregation is unavoidable, because we are using covariates that correspond to each cell. This is an important point. The assumptions we make about the grid will actually effect the way we simulate data. For example, when the only covariate data we have is that which is associated with the quadrat we are working with, then to simulate the data, each point realization will be created in aggregate, that is, a draw from a poisson distribution characterized by an functional of intensity dependent only on the covariates.

If our covariates exist on a finer scale than the quadrat, we can simulate the point locations more finely as well, and take a simulation approach outlined on page 2020 of Banerjee 2014 and followed by Koshkina, 2017. In this approach, we simulate first the total number of observations, create a homogenous poisson process, and then thin the process at the level of the grid of covariates we are working with. It doesn't completely make sense to simulate PPP data in this way here, because our covariates only exist at the level of the aggregated quadrat. That is, we assume that the intensity is constant across our region A, because according to our true data generating process, it is. This is also convenient for the approximation we are using.

We must also take into account the bias associated with the observation of species with true intensity $\lambda(s)$. In order to account for the bias caused by collecting presence-only observations, the poisson process is thinned by the sampling bias b. For more information about thinning a poisson process, see https://math.stackexchange.com/questions/580883/thinning-a-poisson-process.

$$\lambda(s) = \theta(s)b(s) = \exp[\alpha + \beta' x(s) + \gamma + \delta' z(s)]$$

The question of the data generating process under our modelling assumptions is an interesting one. Referencing our initial theoretical outline, we need to simulate points with likelihood $\prod_i \lambda(s_i)\frac{\exp(-\lambda(D))}{n!}$. In the homogenous case, we can simulate n observations from $N(D) \sim Poisson(\lambda|D|)$ ($\lambda(s) = \lambda$ and is thus constant over D), and then according to the n, distribute these uniformly over D. In the case of the Nonhomogenous Poisson Process, we can do the same, drawing n now from $N(D) \sim Poisson(\lambda(D)) = Poisson(\int_D \lambda(s)ds)$, and then distributing their locations over D. Instead of uniformly distributing them however, they are placed according to the distribution $\frac{\lambda(s)}{\lambda(D)}$. There are two issues here, both relating to the evaluation of the integral for the expected number of observations.

Remember that we are using as intensity $\lambda(A_i) = \int_A \lambda(s)ds = \int_A \exp[\alpha + \beta' x(s)]ds = \exp[\alpha + \beta' x_i]$, because we only have information X_i for quadrat A_i centered around s_i. While we

are doing simulations, we could arbitrarily create a finer covariate grid but we won't have access to this during the modelling process. Using the x_i we have, both $\lambda(A)$ and $\lambda(D)$ will be approximations at the covariate resolution, using the assumption that each x_i is constant over A_i. This being the case, then for each observation, we will use the same distribution as the other points that fall into the same A_i. That is, multiple points will have the same distribution $\frac{\exp[\alpha+\beta' x_i]}{\sum_i \exp[\alpha+\beta' x_i]}$. This goes somewhat against the idea of distributing the points in continuous space. Therefore, given the independence of each A_i conditional on the covariates, another approach is to generate counts for each A_i and then uniformly distribute them within each A_i. If x_i is constant across A_i, this is ok. However, if we cannot assume this, using the continuous probability distribution for each point $(\lambda(s)/\lambda(D))$ is necessary to avoid the integral of $\lambda(A)$. We can do so in a way that avoids computing $\lambda(D)$ as well (Banerjee, 2014. p220), by computing λ_{max}, simulating n from $N(D) \sim Poisson(\lambda_{max}|D|)$, and the rejecting some of these points by $\lambda(s_i)/\lambda_{max}$. In this way, the only thing we need is information at the level of $\lambda(s_i)$ and avoid $\lambda(D)$ and $\lambda(A)$.

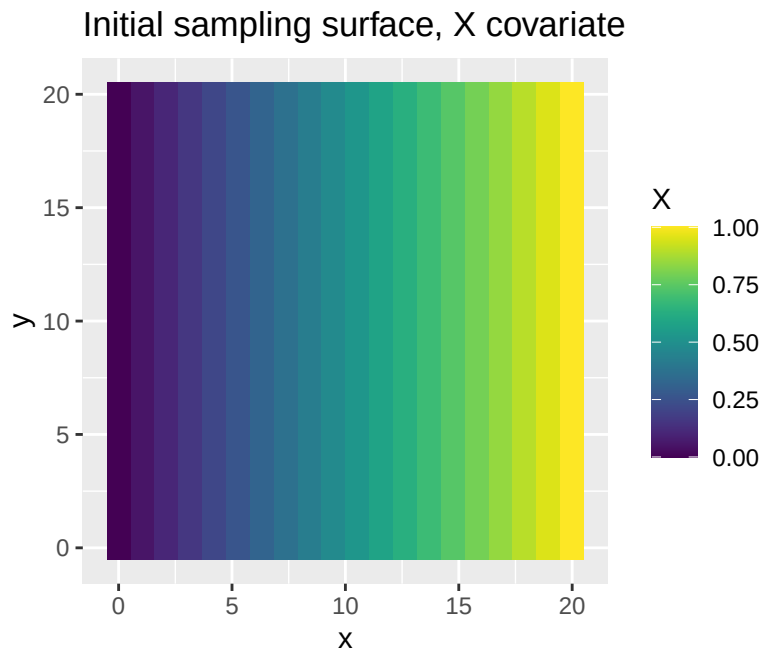
So note that we take the route of simulation depending on our available information. If we only have x_i for each quadrat, directly generating the points as a poisson random variable is a good option because it avoids any complicated integrals. But if we have finer covariates than at the quadrat level we are working with, then thinning the process down with the approach of Banerjee and others is better because it also avoids the integrals and has finer resolution.

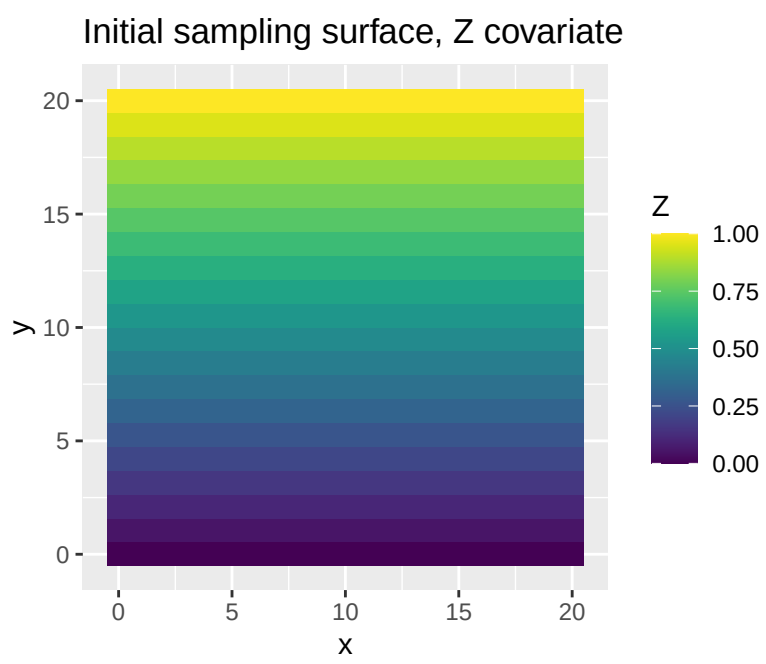
There are a few things to during the data gerating process with the NHPP. The area of the total space, $|D|$, the area of each site (quadrat) $|A_i|$, and the number of cells c that D is discretized into. The area of A depends on the number of discretizations (or cells): $|A_i| = |D|/c$. The intensity will be scaled by this factor $|D|/c$, so that as the area of $|A|$ becomes smaller, the intensity converges to that of a "true" continuous poisson process with its center at A_i. We also assume that the intensity is constant over A, so that we can indeed scale the intensity by the area factor, instead of needing to integrate over A. This assumption is in contrast to assuming that we model the instensity directly as $\Lambda(A_i) = \exp(\beta' X)$ (versus $\Lambda(A_i) = \frac{|D|}{c}\exp(\beta' X)$).

Thi following code generates observations at each site and plots observed intensities, bias, and counts.

```
area_D <- 100
k <- 20
sites <- k^2
# Parameter values somewhere around Fithian 2015.
alpha <- -2
beta <- 2
gamma <- -2
delta <- 0.5
params <- list(alpha=alpha, beta=beta, gamma=gamma, delta=delta)
# Plot all intensity functions
```

```r
grid_points <- get_sampling_surface_simple(k)
```





```r
distance_mat <- as.matrix(dist(grid_points[,1:2]))
data_reps <- 1
corr_matrix <- specify_corr(grid_points[,1:2])
gp_bool <- F
sim_data <- generate_ppp_data_r(grid_points, params, sites, data_reps, corr_matrix, gp_boo
Y_positive_indices <- which(sim_data$Y>0)
# Check that total counts equal number of coordinates
stopifnot(sum(sim_data$Y[Y_positive_indices]) == nrow(sim_data$coordinates))
# This is the data and coordinates at which there are counts > 0
sim_data$Y[Y_positive_indices]
```

```
[1] 1 1 1 1 1 1
```

```r
sim_data$Y_coords
```

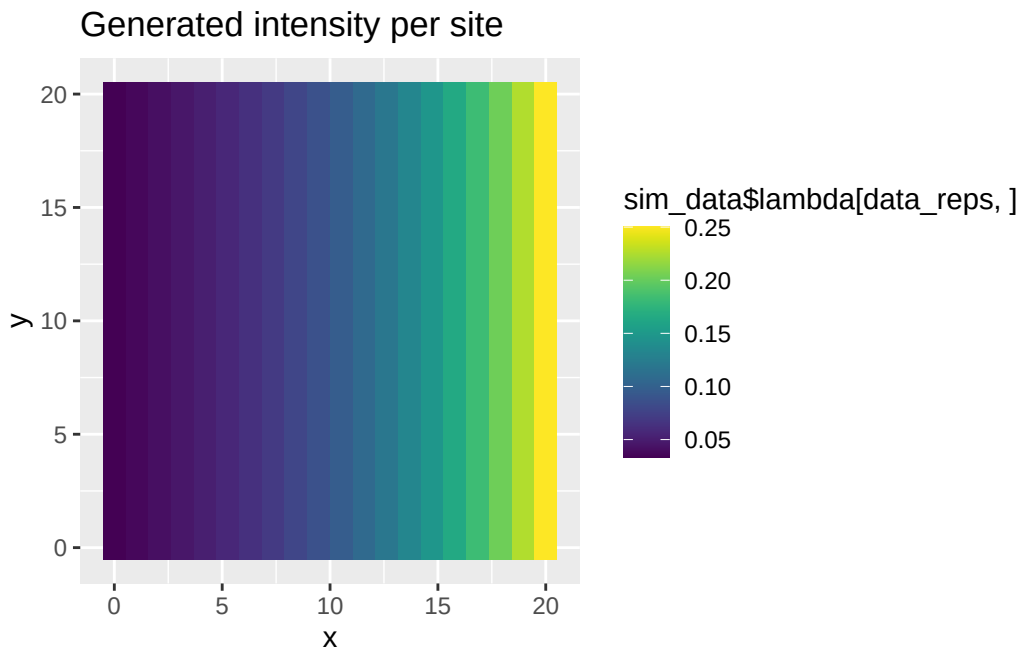
```
          x         y
1 16.025076  3.023165
2 20.187234  4.505764
3 16.140415  6.322235
4  3.597623  9.827059
5  9.451658 12.727483
6 17.517565 17.053193
```

```r
grid_points[Y_positive_indices,]
```

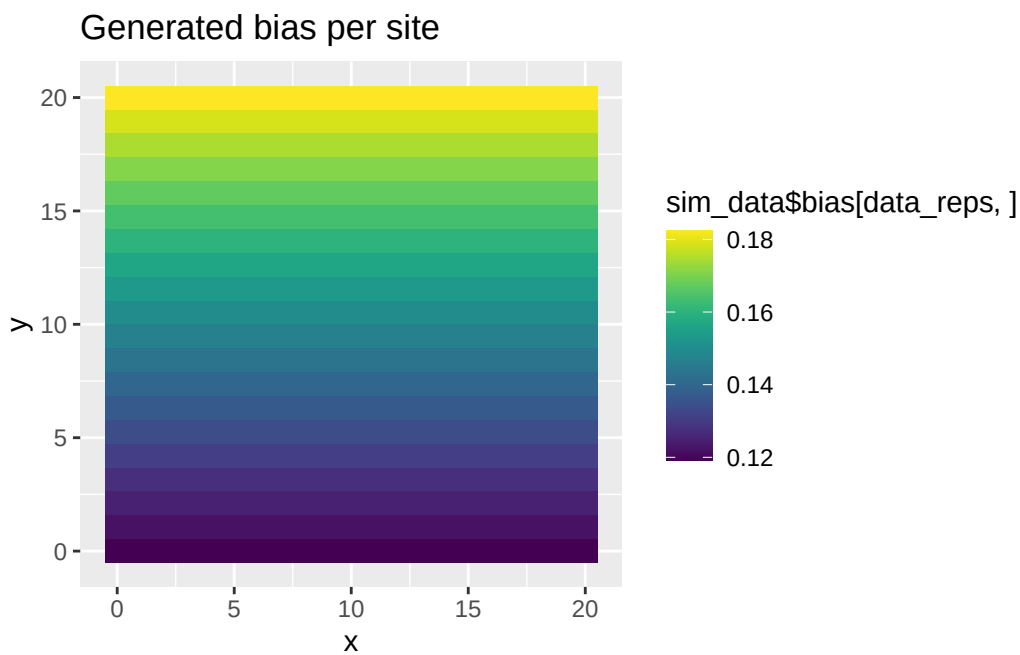
```
            x         y     aux_x     aux_z
76  15.789474  3.157895 0.7894737 0.1578947
100 20.000000  4.210526 1.0000000 0.2105263
136 15.789474  6.315789 0.7894737 0.3157895
184  3.157895  9.473684 0.1578947 0.4736842
250  9.473684 12.631579 0.4736842 0.6315789
338 17.894737 16.842105 0.8947368 0.8421053
```

```r
# Use the final generation iteration
p <- ggplot(grid_points, aes(x, y, fill=sim_data$lambda[data_reps,])) +
  geom_tile() +
  scale_fill_viridis(discrete=FALSE) +
  ggtitle("Generated intensity per site")
```

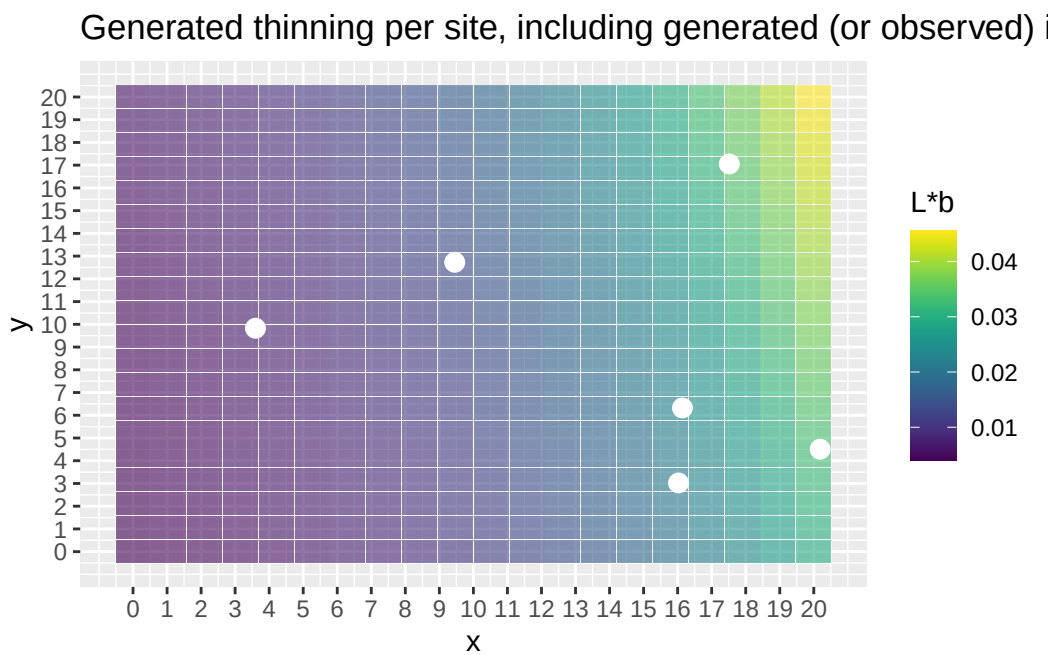
```r
print(p)
```



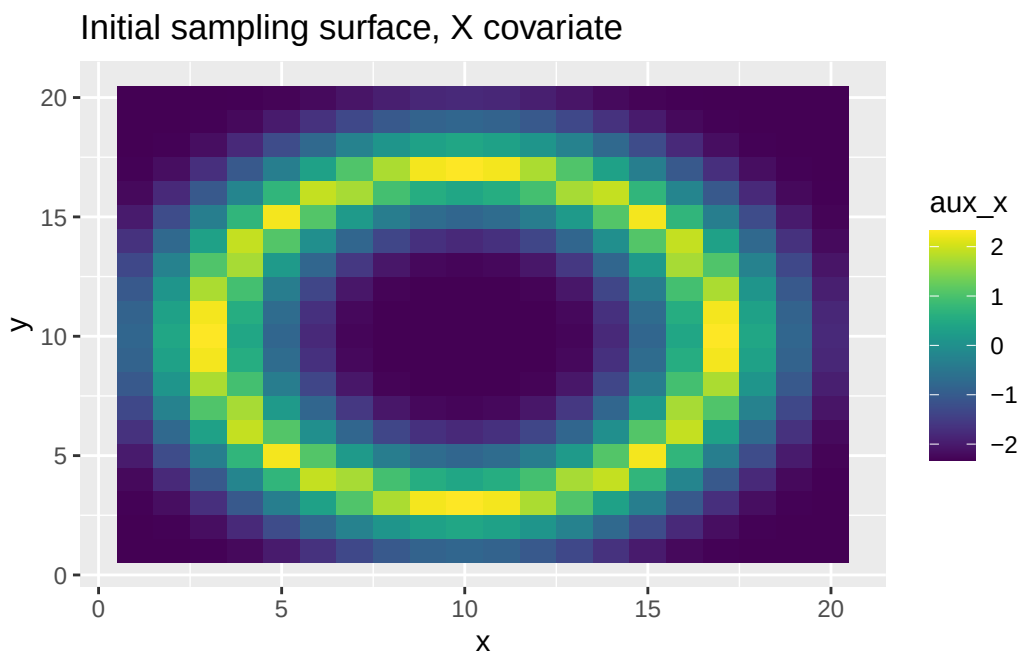

```r
p <- ggplot(grid_points, aes(x, y, fill=sim_data$bias[data_reps,])) +
  geom_tile() +
  scale_fill_viridis(discrete=FALSE) +
  ggtitle("Generated bias per site")
print(p)
```




```
thinned_intensity <- sim_data$lambda[data_reps,]*sim_data$bias[data_reps,]
p <- ggplot() +
  geom_tile(grid_points, mapping=aes(x, y, fill=thinned_intensity, width=1, height=1), alp
  scale_fill_viridis(discrete=FALSE, name="L*b") +
  ggtitle("Generated thinning per site, including generated (or observed) individuals") +
  geom_point(data=sim_data$Y_coords, mapping=aes(x=x, y=y), size=3, col="white") +
  theme(panel.grid.minor = element_line(colour="white")) +
  scale_y_continuous(breaks = seq(0, 20, 1)) +
  scale_x_continuous(breaks = seq(0, 20, 1))
print(p)
```

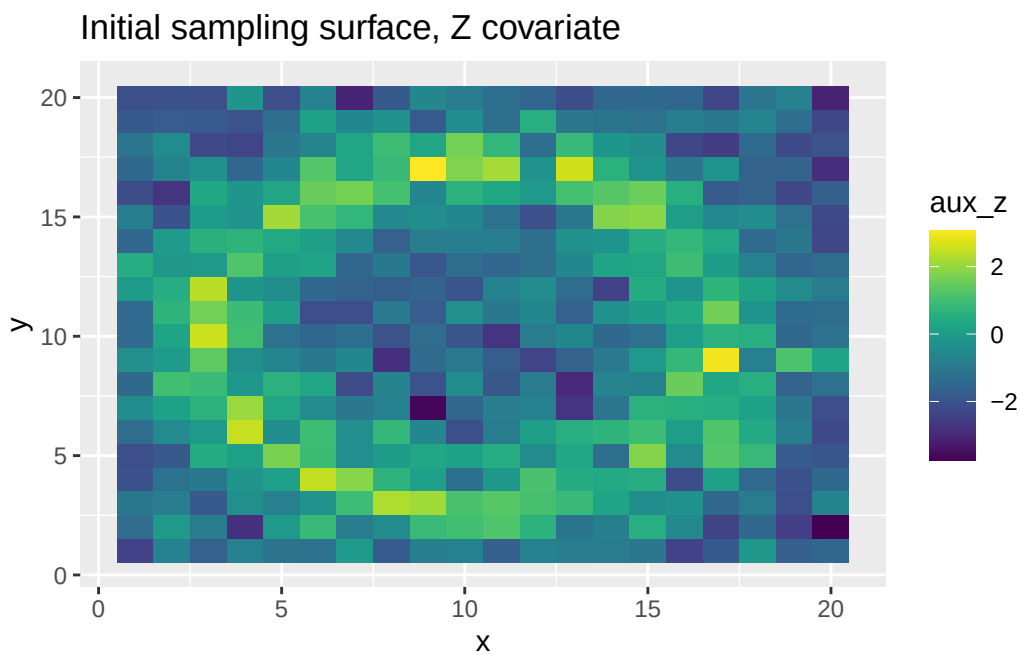



```
# Then use the donut intensity with correlated bias
aux_cor <- 0.7
grid_points <- get_sampling_surface_donut(k)
```




```
grid_points <- get_bias_surface_correlated(grid_points, aux_cor)
```

```
[1] "Correlation between X and Z"
          [,1]      [,2]
[1,] 1.0000000 0.8009806
[2,] 0.8009806 1.0000000
```




```
sites <- sites/4
grid_points <- grid_points %>% dplyr::filter(x<(k/2)+1, y<(k/2)+1)
# Filter for only a quarter of the grid.
# If we filter, we need to change the total sites as well
# Need distances for LGCP
distance_mat <- as.matrix(dist(grid_points[,1:2]))
data_reps <- 1
corr_matrix <- specify_corr(grid_points[,1:2])
gp_bool <- F
sim_data <- generate_ppp_data_r(grid_points, params, sites, data_reps, corr_matrix, gp_boo
Y_positive_indices <- which(sim_data$Y>0)
stopifnot(sum(sim_data$Y[Y_positive_indices]) == nrow(sim_data$coordinates))
sim_data$Y[Y_positive_indices]
```

```
 [1] 3 1 3 1 3 4 1 1 5 6
```

```
sim_data$Y_coords
```

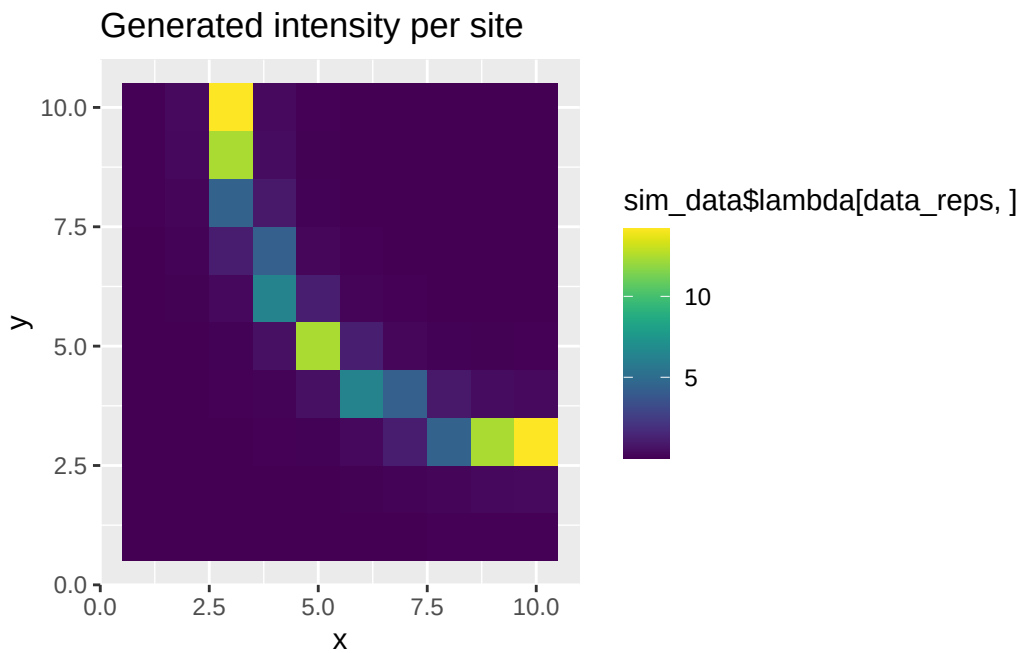
```
          x         y
1  7.742985  3.285383
```

```
2  7.520093  2.745805
3  8.434541  2.629894
4  8.707367  2.754918
5  9.914455  2.501376
6  9.709449  2.658010
7  9.615209  3.125861
8  5.843431  3.780790
9  6.688967  3.989234
10 6.770086  3.887112
11 7.032844  3.512671
12 4.811001  5.137054
13 5.436905  4.755995
14 5.086746  5.480601
15 4.578128  4.724297
16 4.174519  6.292339
17 4.315530  6.642341
18 3.486645  9.282819
19 2.521810  9.296045
20 3.134821  8.670107
21 2.899662  8.514156
22 2.845044  8.516232
23 2.840559  9.630771
24 2.828090 10.470985
25 2.953504  9.932560
26 3.157144 10.460068
27 3.219651  9.799521
28 2.828509 10.459985
```

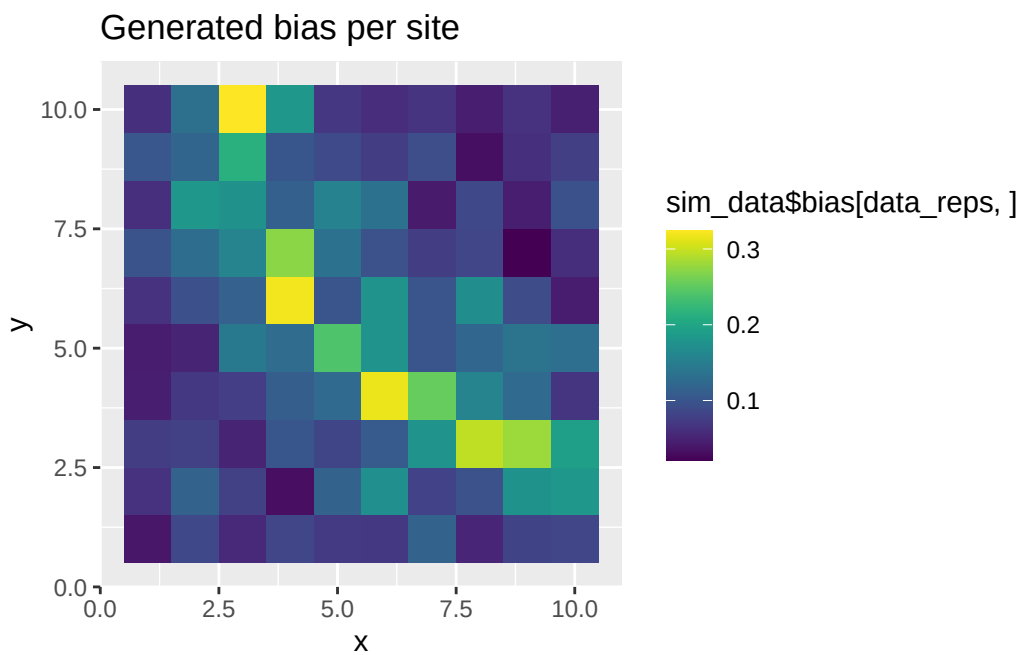
```
grid_points[Y_positive_indices,]
```

```
    x  y    aux_x    aux_z
28  8  3 1.747454 2.261237
29  9  3 2.257839 2.109223
30 10  3 2.326348 1.108851
36  6  4 1.921817 2.460005
37  7  4 1.718916 1.843953
45  5  5 2.257839 1.694453
54  4  6 1.921817 2.484141
64  4  7 1.718916 2.031675
83  3  9 2.257839 1.391192
93  3 10 2.326348 2.531949
```

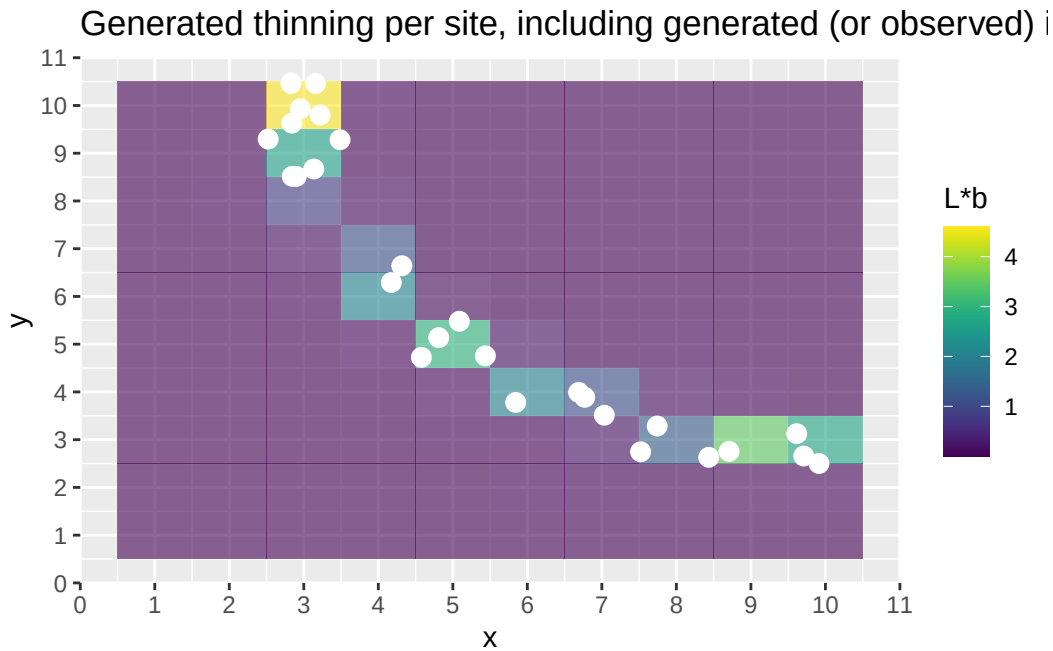
```r
# Use the final generation iteration
p <- ggplot(grid_points, aes(x, y, fill=sim_data$lambda[data_reps,])) +
  geom_tile() +
  scale_fill_viridis(discrete=FALSE) +
  ggtitle("Generated intensity per site")
print(p)
```




```r
p <- ggplot(grid_points, aes(x, y, fill=sim_data$bias[data_reps,])) +
  geom_tile() +
  scale_fill_viridis(discrete=FALSE) +
  ggtitle("Generated bias per site")
print(p)
```




```
thinned_intensity <- sim_data$lambda[data_reps,]*sim_data$bias[data_reps,]
p <- ggplot() +
  geom_tile(grid_points, mapping=aes(x, y, fill=thinned_intensity, width=1, height=1), alp
  scale_fill_viridis(discrete=FALSE, name="L*b") +
  ggtitle("Generated thinning per site, including generated (or observed) individuals") +
  geom_point(data=sim_data$Y_coords, mapping=aes(x=x, y=y), size=3, col="white") +
  theme(panel.grid.minor = element_line(colour="white")) +
  scale_y_continuous(breaks = seq(0, 20, 1)) +
  scale_x_continuous(breaks = seq(0, 20, 1))
print(p)
```



Next we fit the Thinned Nonhomogenous Poisson Process model in Stan. We can make a few observations about this model. First, examine the likelihood:

$$L(\lambda|Y) = \prod_{i=1}^{N} \frac{\lambda_i^{n_i} \exp[-\lambda_i]}{n_i!}$$

. This is made using the assumption that the counts at sites s_i are distributed following a poisson process with mean λ. This assumption allows for a simpler specification of the likelihood, since we can use the poisson distribution. The setting of $|A| = 1$ allows us to make this assumption, as we do not need to multiple the intensity parameter λ by the area in the model.

The next sections of code fitting and diagnosing the poisson process model in Stan, using the generated data.

```
model_string <- stan_poisson_process
write(model_string, model_path)
# data_reps is 1 here so just use the first index
data = list(N = sites, X = grid_points$aux_x, Z = grid_points$aux_z, y = sim_data$Y[data_r
model <- cmdstan_model(model_path)
fit <- model$sample(data=data, seed=13, chains=3, iter_sampling=10000, iter_warmup=1000, t
```

```
fit$summary(variables=c('alpha', 'beta', 'gamma', 'delta'))
```

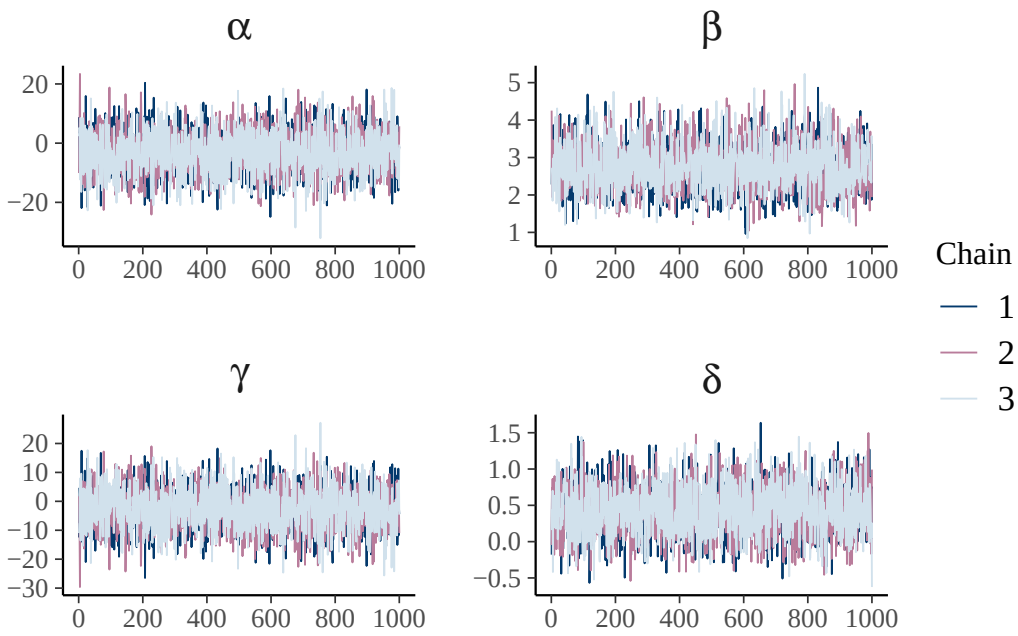
```
# A tibble: 4 x 10
  variable   mean median    sd   mad      q5   q95  rhat ess_bulk ess_tail
  <chr>     <dbl>  <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl>    <dbl>    <dbl>
1 alpha    -2.93  -2.96  7.14  7.32  -14.7    8.78  1.00    3106.    2681.
2 beta      2.75   2.71  0.622 0.606   1.78   3.84  1.00    2833.    2741.
3 gamma    -2.80  -2.86  7.16  7.26  -14.4    9.13  1.00    3071.    2680.
4 delta     0.427  0.410 0.339 0.337  -0.122  1.00  1.00    2950.    2866.
```

From the table above, we can observe that the parameters β and δ have been recovered. However, due to the correlation of α and γ, only the sum $\alpha + \gamma$ can be correctly identified. Our "true" sum of was equal to -6, and we can see the estimate here is -5.5, an estimate which also incorporates prior information. The correlation between the two can be observed in the scatter plots below, as well as in the wide Confidence Interval of the posterior, relative to the other parameters.

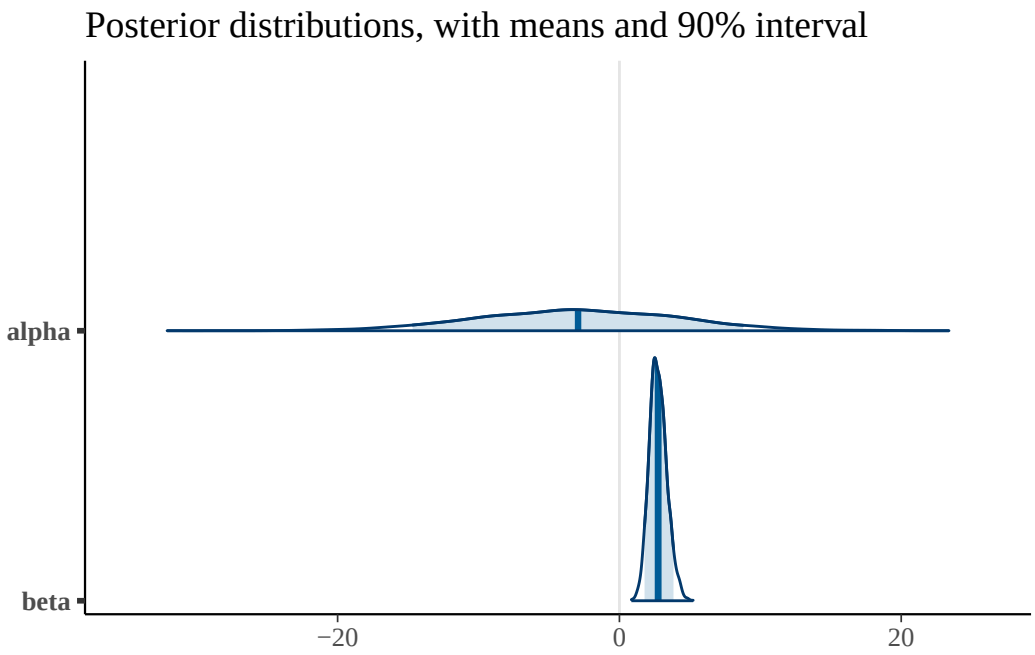
2.2 Diagnostics

Here we check goodness of fit and PPCs. We also discuss some advantages and disadvantages of PPCs. One thing I'm still interested in is goodness of fit and PPCs for species distribution models?

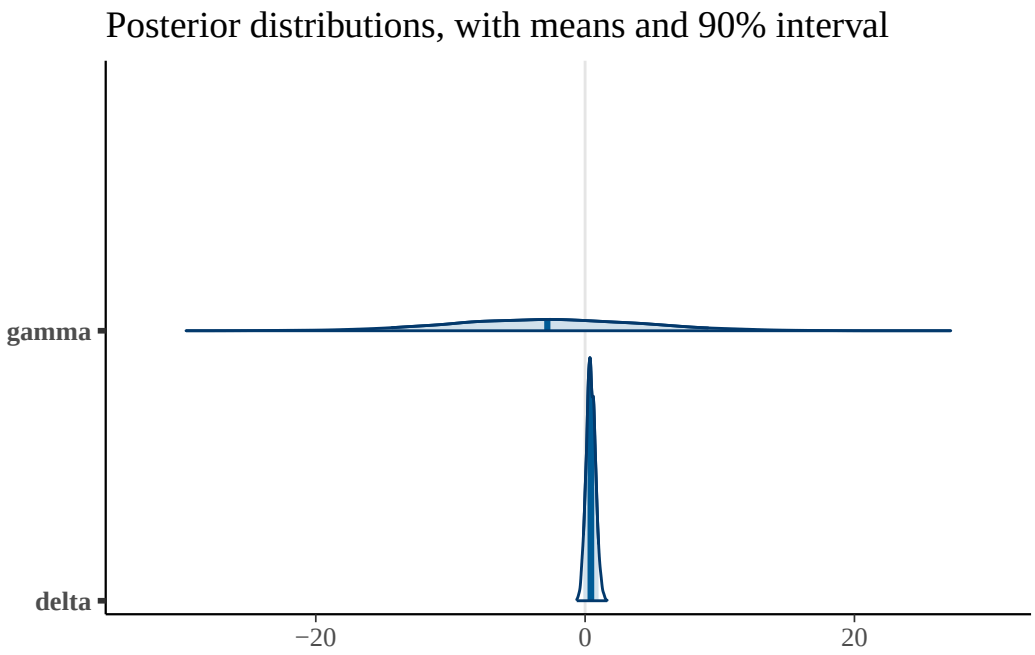
```r
posterior <- fit$draws()
color_scheme_set("mix-blue-pink")
p_trace <- mcmc_trace(posterior,  pars = c("alpha", "beta", "gamma", "delta"),
                      facet_args = list(nrow = 2, labeller = label_parsed))
print(p_trace + facet_text(size = 15))
```



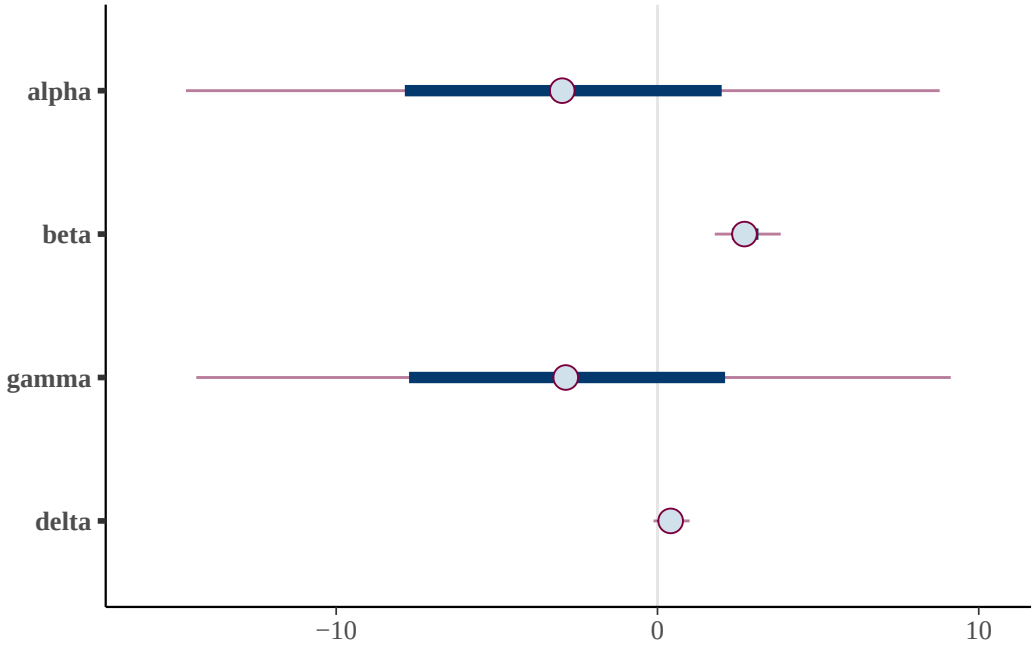

```
plot_title <- ggtitle(paste("Posterior distributions, with means and 90% interval"))
p_post <- mcmc_areas(posterior,  prob = 0.9, point_est="mean", regex_pars = c("alpha", "be
print(p_post)
```



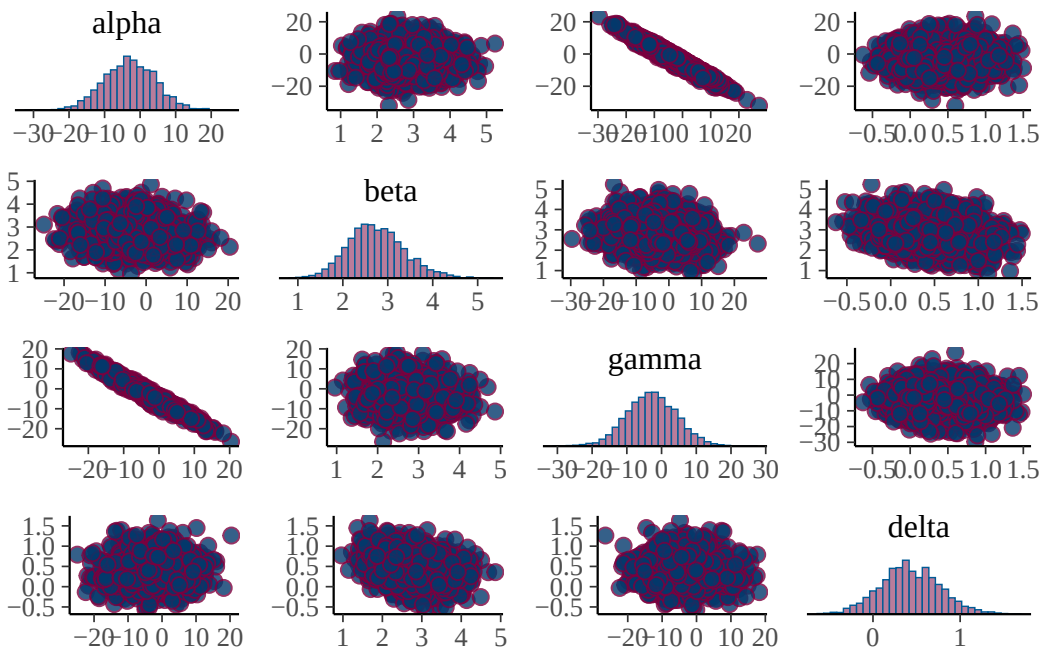

```
plot_title <- ggtitle(paste("Posterior distributions, with means and 90% interval"))
p_post <- mcmc_areas(posterior,  prob = 0.9, point_est="mean", regex_pars = c("gamma", "de
print(p_post)
```



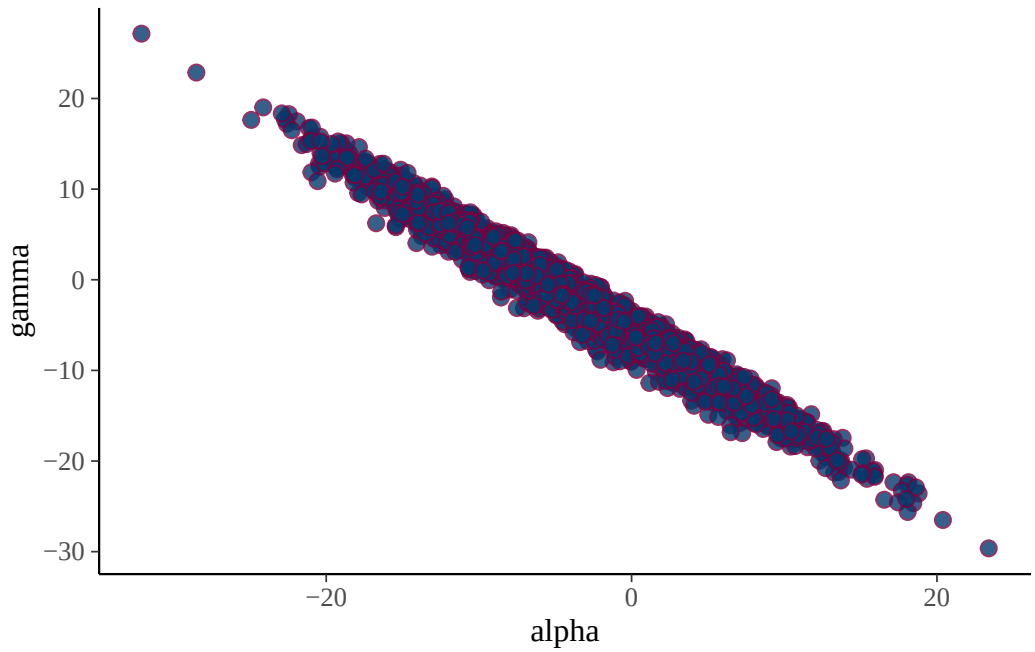

```
mcmc_intervals(fit$draws(), pars=c('alpha', 'beta', 'gamma', 'delta'))
```



```r
#mcmc_hist(fit$draws(), pars = c('alpha', 'beta', 'gamma', 'delta'))
plot_title <- ggtitle("Parameter posterior sample correlation")
p_post <- mcmc_pairs(fit$draws(), pars=c('alpha', 'beta', 'gamma', 'delta'))
print(p_post)
```




```r
mcmc_scatter(fit$draws(), pars=c('alpha','gamma'))
```



We can note the wide intervals on α and γ. Additionally the scatter plot shows the independence of β and δ with the other parameters, as well as the correlation between α and γ.

2.3 PPD

We next look at the Posterior Predictive Distribution (PPD) of the observations per site. That is, we plot the generated counts Y_i for site s_i on the sampling surface (grid).

The PPD can be expressed as the samples from the predictive distribution of Y taking into account the uncertainty surrounding the parameters. The PPD can be written as

$$p(\tilde{y}|Y) = \int p(\tilde{y}|\lambda, b, Y)p(\lambda, b|Y)dbd\lambda$$

such that the posterior uncertainty of the parameters is integrated out.

Below we plot the generated values of $\tilde{y}$ for each site. Additionally, the generated quantities of intensity, bias, and intensity*bias ($ b) per site are also plotted on the grid.

```
generated_vars <- c('lambda_bias', 'lambda_rep', 'b_rep', 'y_rep')
lambda_thinned <- fit$summary(variables=generated_vars[1])$mean
```

```r
lambda_rep <- fit$summary(variables=generated_vars[2])$mean
bias_rep <- fit$summary(variables=generated_vars[3])$mean
y_ppd <- fit$summary(variables=generated_vars[4])$mean
ppd_df <- data.frame(x=grid_points$x, y=grid_points$y, y_rep=y_ppd, lt=lambda_thinned, l=l
head(ppd_df)
```

```
  x y y_rep           lt        l         b
1 1 1     0 2.703176e-05 15170.15  50721151
2 2 1     0 5.028806e-05 15220.90 137665637
3 3 1     0 3.535569e-05 15768.22  79018114
4 4 1     0 5.810691e-05 19318.27 133352130
5 5 1     0 7.520168e-05 34260.40 106139019
6 6 1     0 1.406662e-04 79838.86 104236442
```

It is important to notice that while the thinned intensity is ostensibly realistic in estimation, the intensity and bias are not. This is due to the specification of the Thinned Poisson Process

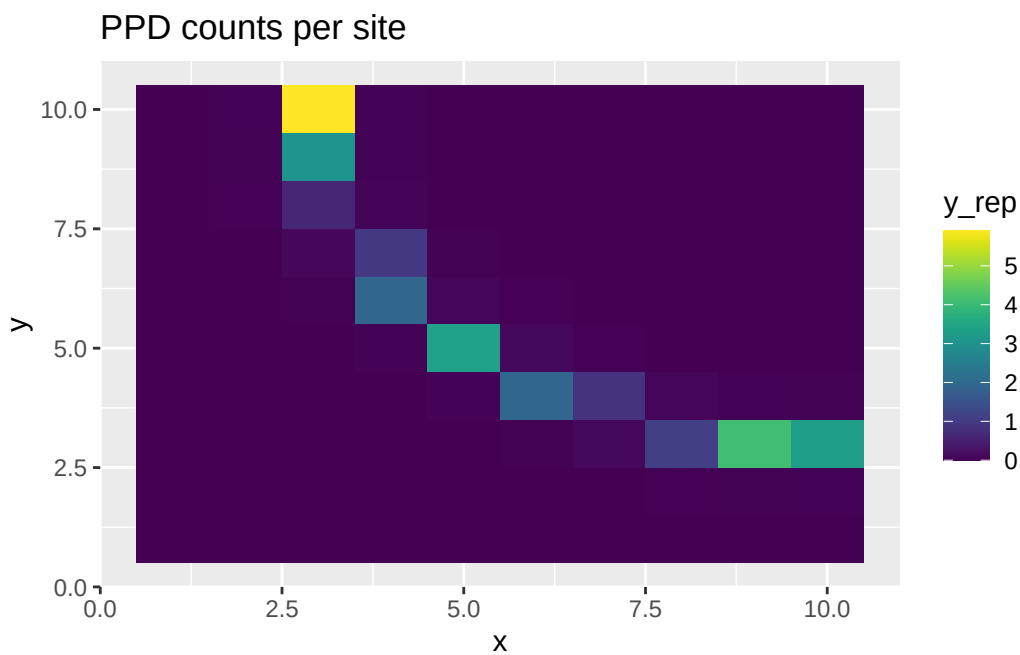
$$\lambda^* = \lambda(s)b(s)$$

such that

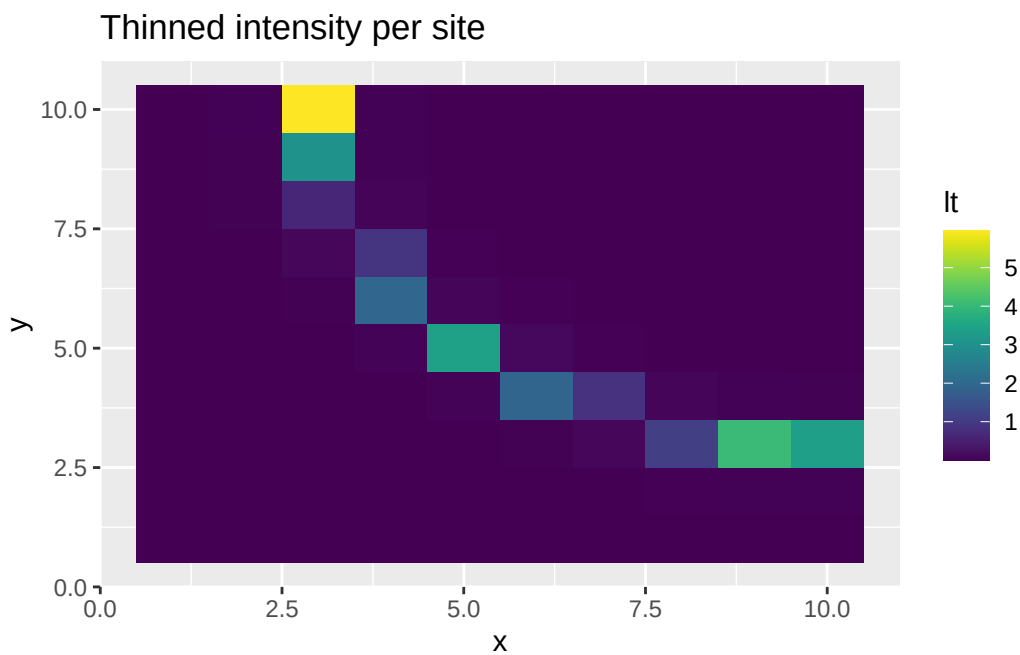
$$\lambda^* = \exp\left[\alpha + \beta * X(s) + \gamma + \delta * Z(s)\right]$$

We can then see the generated data in the sampling surface:

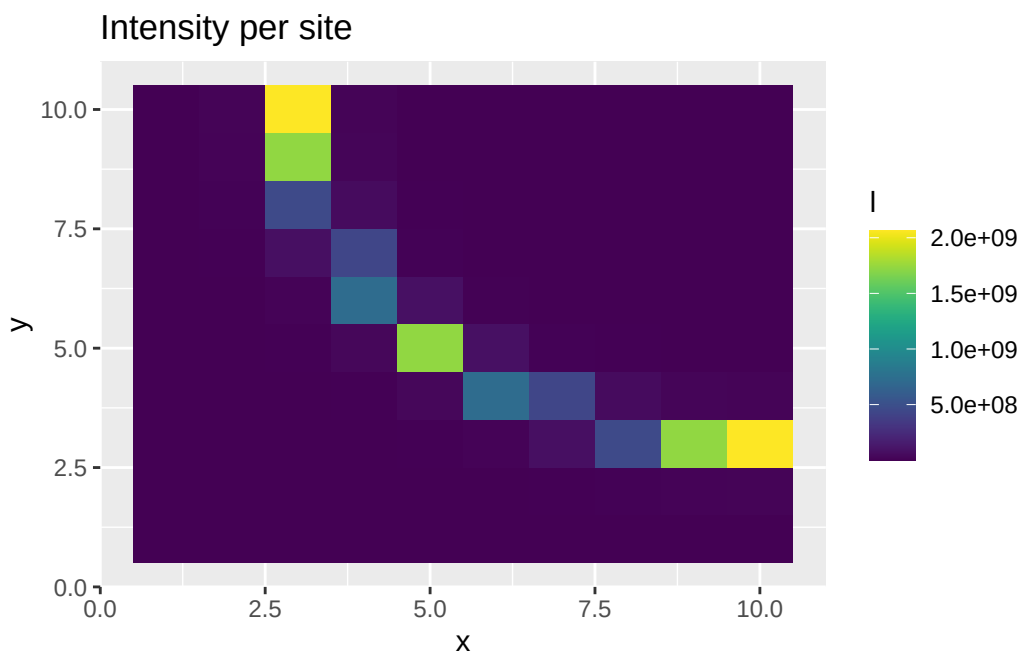
```r
p <- ggplot(ppd_df, aes(x, y, fill=y_rep)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("PPD counts per site")
print(p)
```



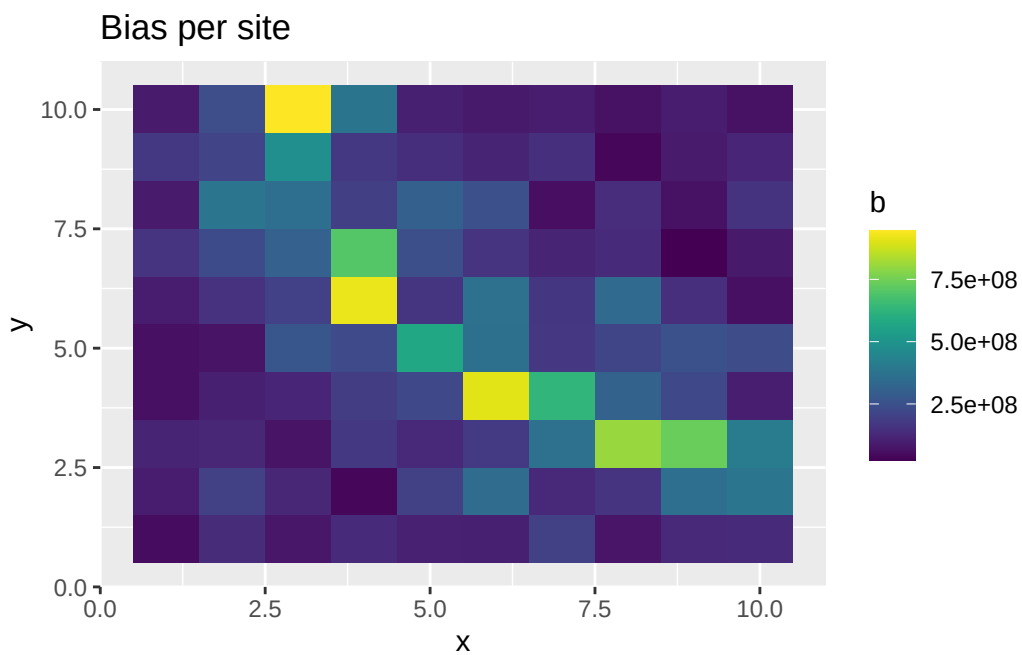

```
p <- ggplot(ppd_df, aes(x, y, fill=lt)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Thinned intensity per site")
print(p)
```




```
p <- ggplot(ppd_df, aes(x, y, fill=l)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Intensity per site")
print(p)
```




```
p <- ggplot(ppd_df, aes(x, y, fill=b)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Bias per site")
print(p)
```

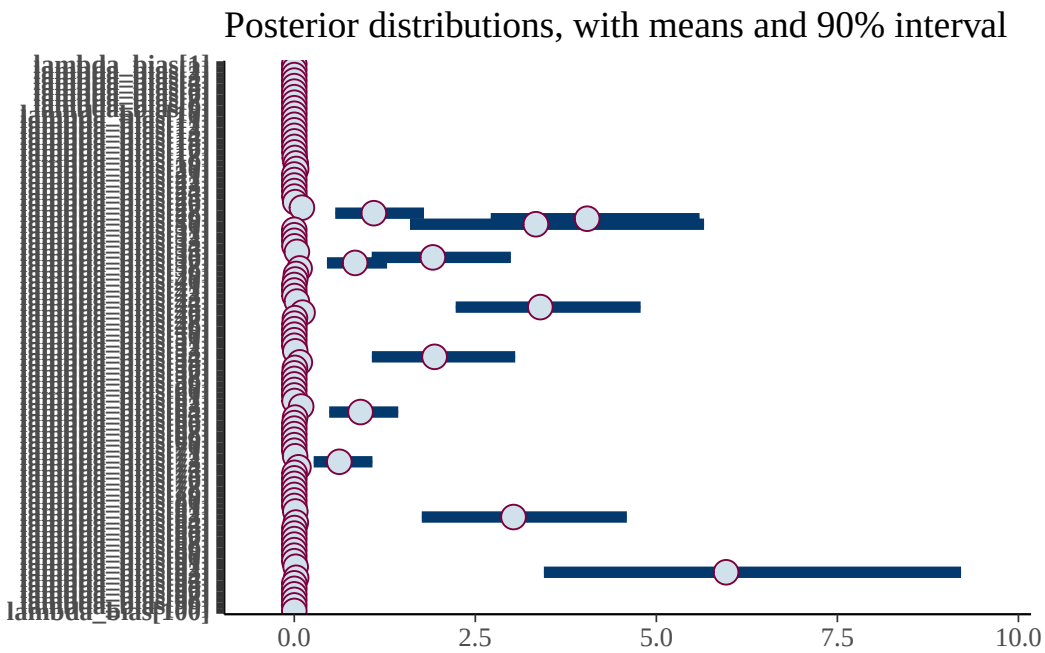


2.4 Correlation between coefficients in Thinned poisson process model for presence-only data.

In the above plots, we can notice that the bias and (unthinned) intensity are behaving strangely. Here we observe the effect of correlated parameters describing the behavior of the intensity and the bias the NHPP. We examine the distributions of the generated quantities using the fitted model.

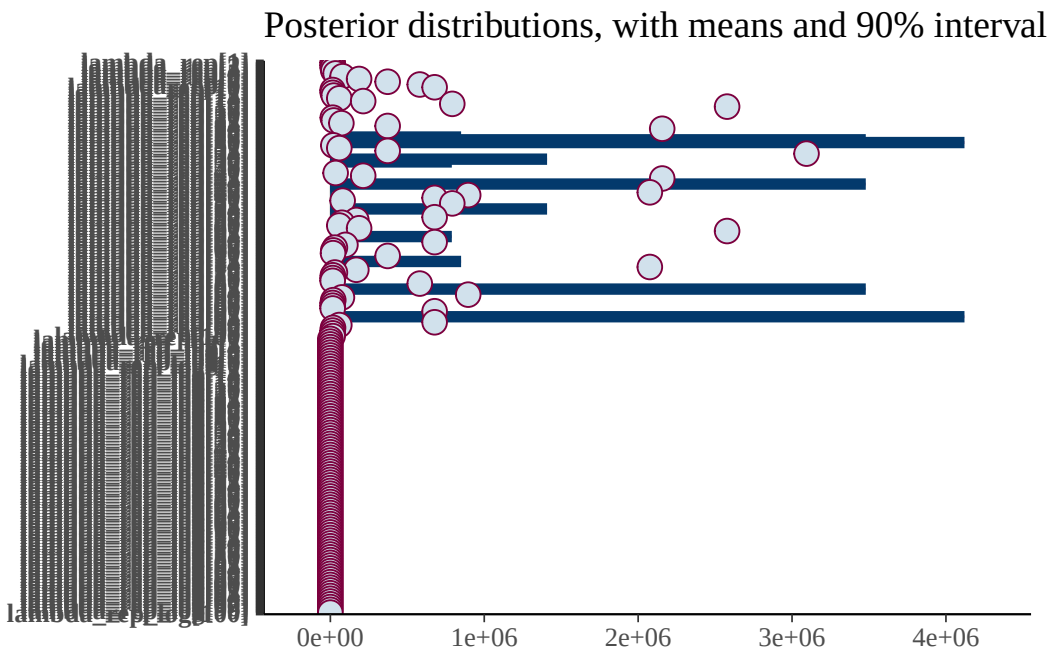
The issue with the large values is both because we are taking the mean of the posterior distribution as our point estimate, and the high correlation between the two parameters causes the estimates to inversely fluctuate around each other.

```
plot_title <- ggtitle(paste("Posterior distributions, with means and 90% interval"))
p_post <- mcmc_intervals(posterior,  prob = 0.9, point_est="mean", regex_pars = generated_
print(p_post)
```



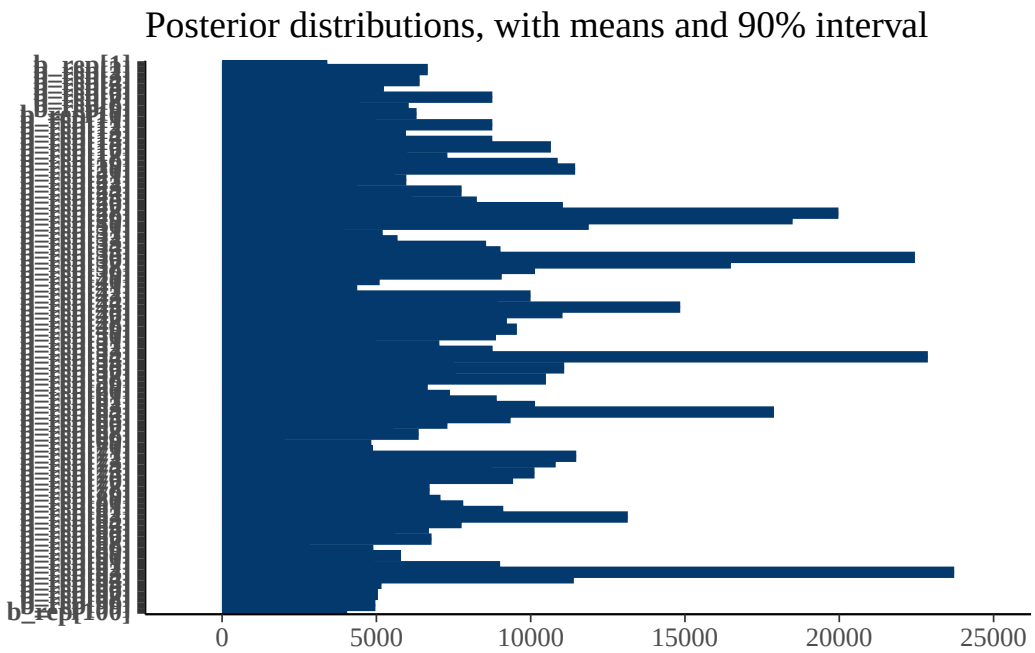

```
p_post <- mcmc_intervals(posterior,  prob = 0.9, point_est="mean", regex_pars = generated_
print(p_post)
```

```
Warning: Removed 34 rows containing missing values (`geom_point()`).
```



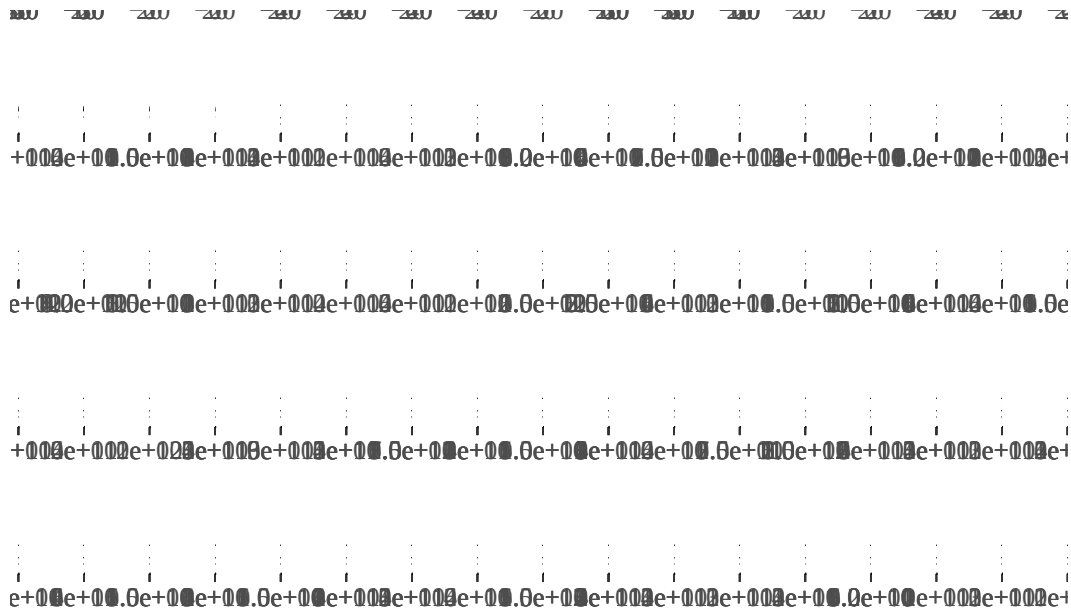

```
p_post <- mcmc_intervals(posterior,  prob = 0.9, point_est="mean", regex_pars = generated_
print(p_post)
```

```
Warning: Removed 100 rows containing missing values (`geom_point()`).
```



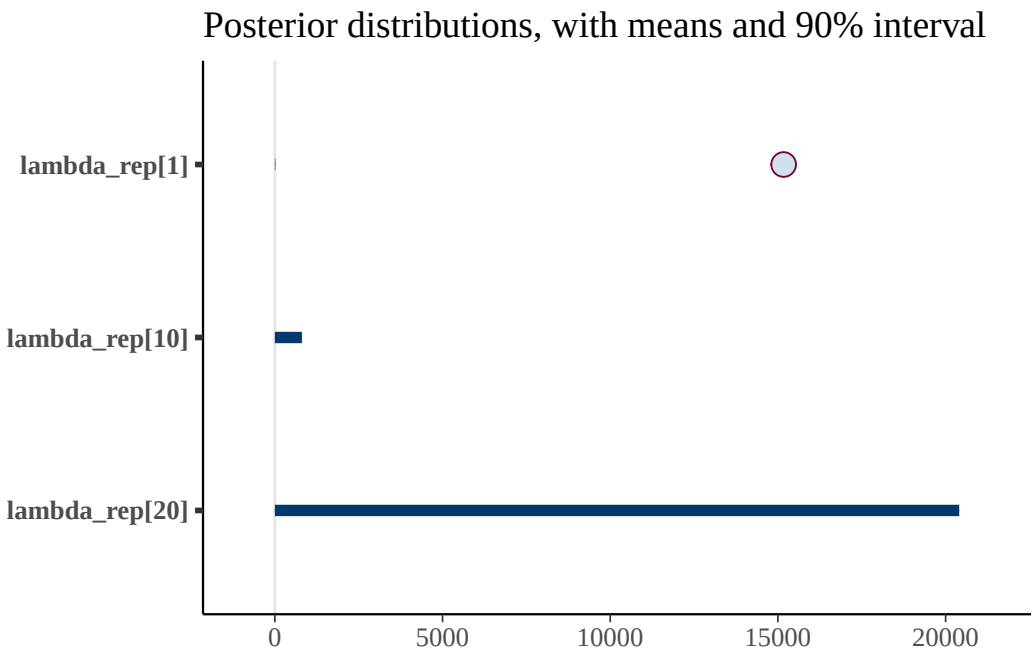

```
# Look at counts for each site
gen_y <- c("y_rep[1]", "y_rep[10]", "y_rep[20]")
p_post <- mcmc_hist(posterior, pars=)
print(p_post)
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



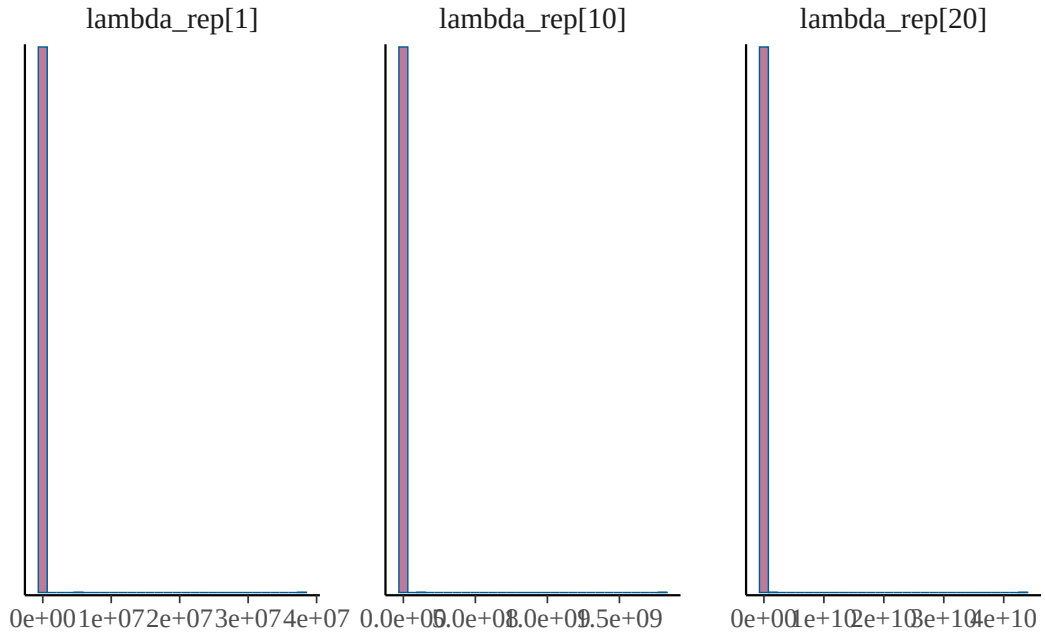
```r
# The gen parameters for just a few of the sites
gen_lambda <- c('lambda_rep[1]', 'lambda_rep[10]', 'lambda_rep[20]')
gen_lb <- c('lambda_bias[1]', 'lambda_bias[10]', 'lambda_bias[20]')
gen_b <- c('b_rep[1]', 'b_rep[10]', 'b_rep[20]')
p_post <- mcmc_intervals(posterior,  prob = 0.9, point_est="mean", pars = gen_lambda) + pl
print(p_post)
```

```
Warning: Removed 2 rows containing missing values (`geom_point()`).
```



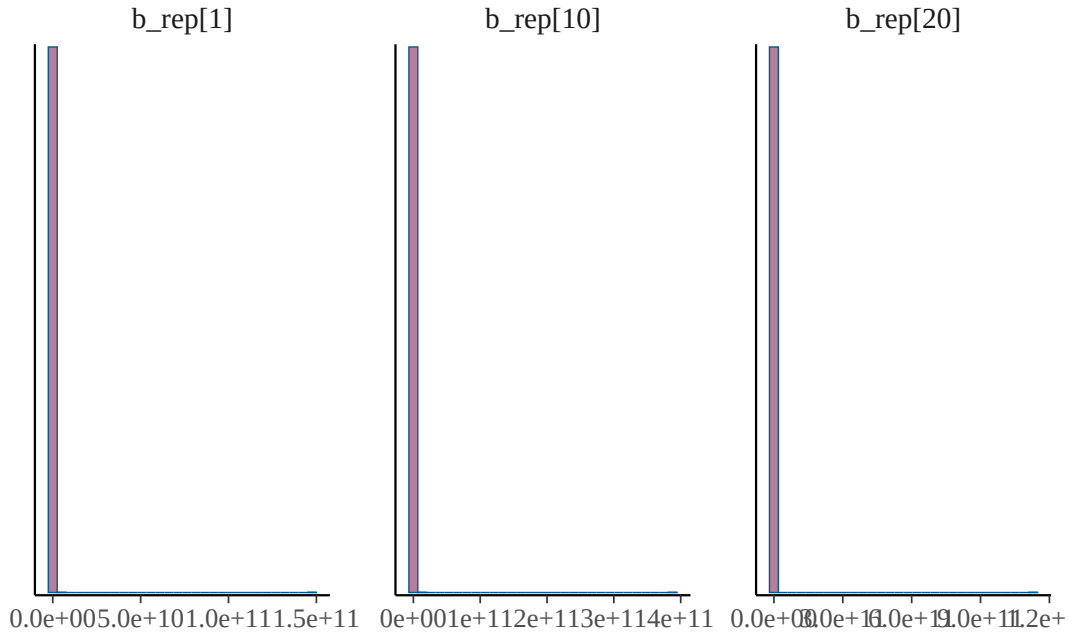

```
p_post <- mcmc_hist(posterior, pars = gen_lambda)
print(p_post)
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



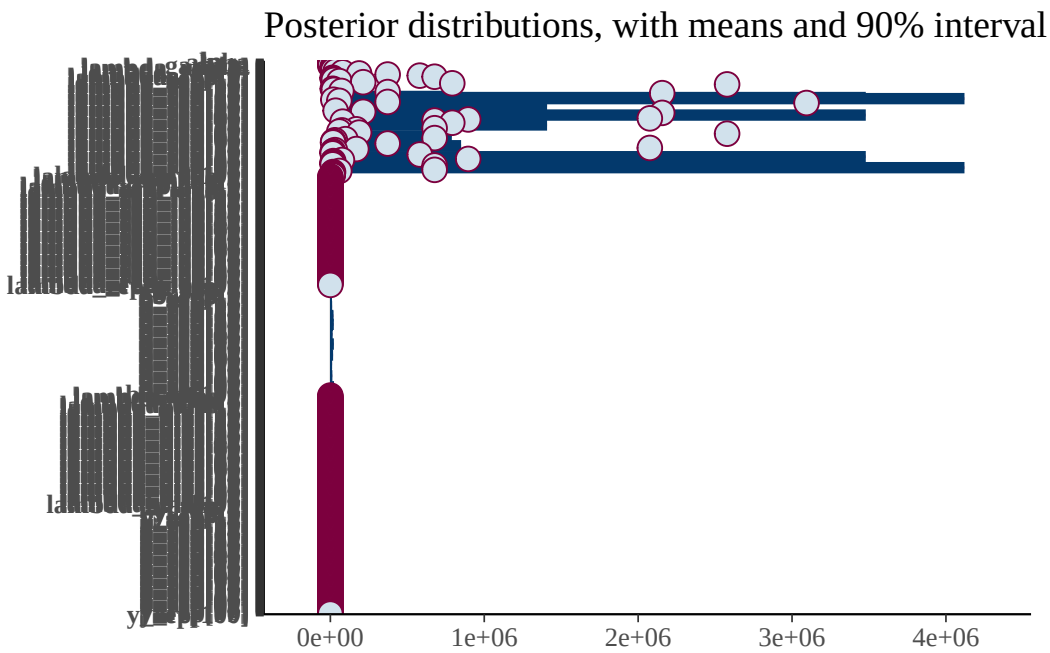

```
p_post <- mcmc_hist(posterior, pars = gen_b)
print(p_post)
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
p_post <- mcmc_intervals(posterior,  prob = 0.9, point_est="mean", pars = ) + plot_title
print(p_post)
```

```
Warning: Removed 134 rows containing missing values (`geom_point()`).
```

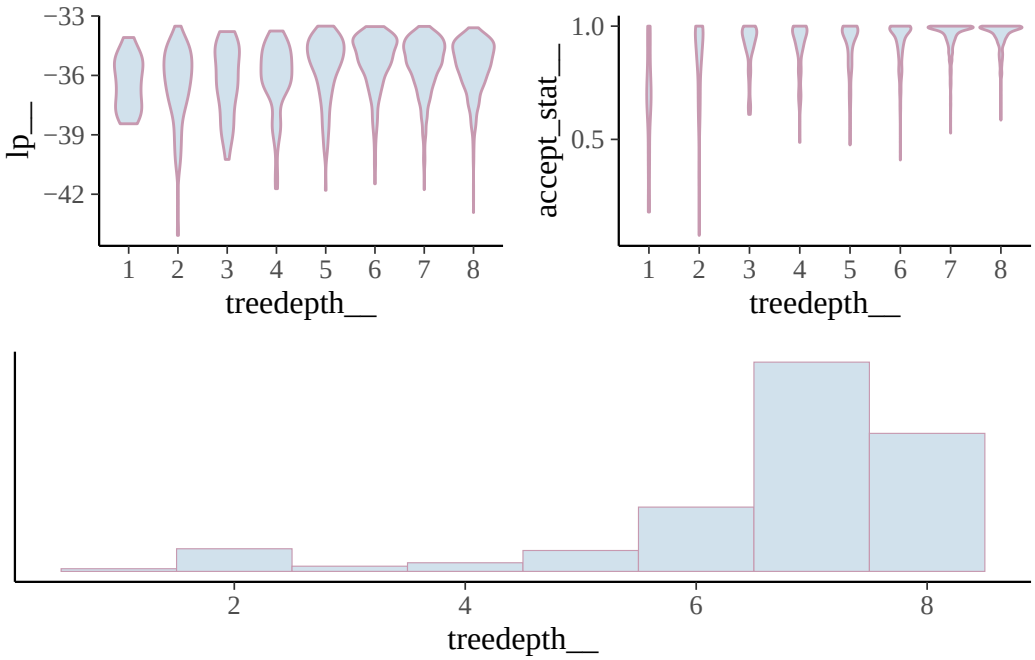


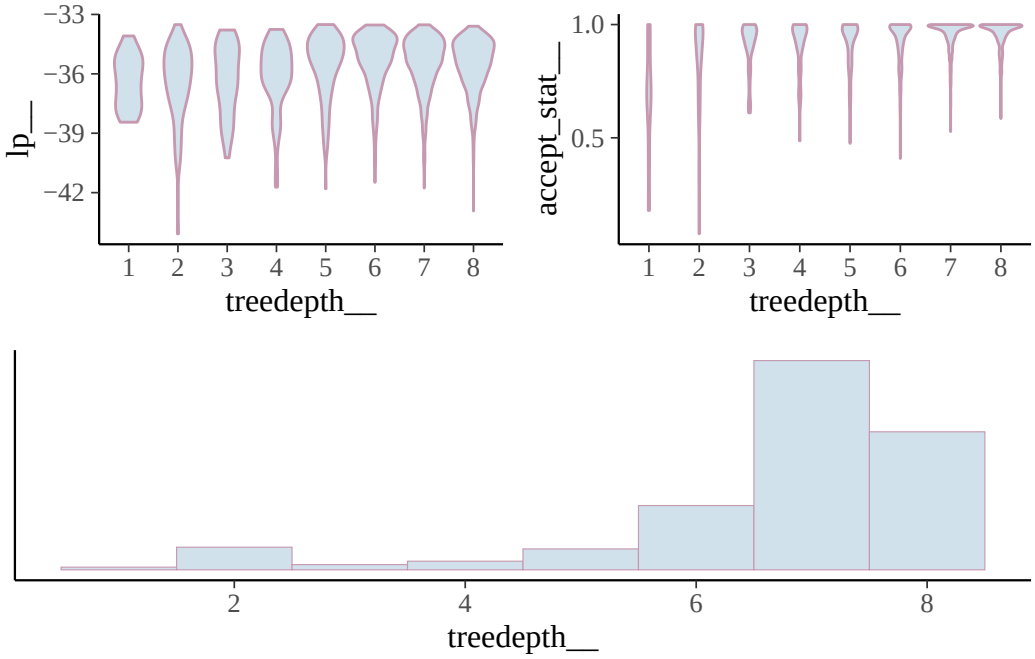
The take away is that the mean for the generated quantities is very skewed by the occassional large value. This high skew is caused by the correlation between b and l, since these parameters are not estimable between themselves. Therefore, we arrive at reasonable estimates for b*l since this is draw specific (where when one is high the other is low), but the overall mean for either b or l over multiple draws cannot be trusted. This is clear from the histograms of λ and the bias, where most of the values take on very low values, but do fluctuate higher. And when one goes higher, the other goes lower, which I'd like to check a specific example of, for one draw.

```
# Nuts diagnostics that I haven't tried to understand really
nuts_fit <- nuts_params(fit)
head(nuts_fit)
```

```
  Chain Iteration   Parameter Value
1     1         1 treedepth__     6
2     1         2 treedepth__     7
3     1         3 treedepth__     1
4     1         4 treedepth__     7
5     1         5 treedepth__     7
6     1         6 treedepth__     7
```

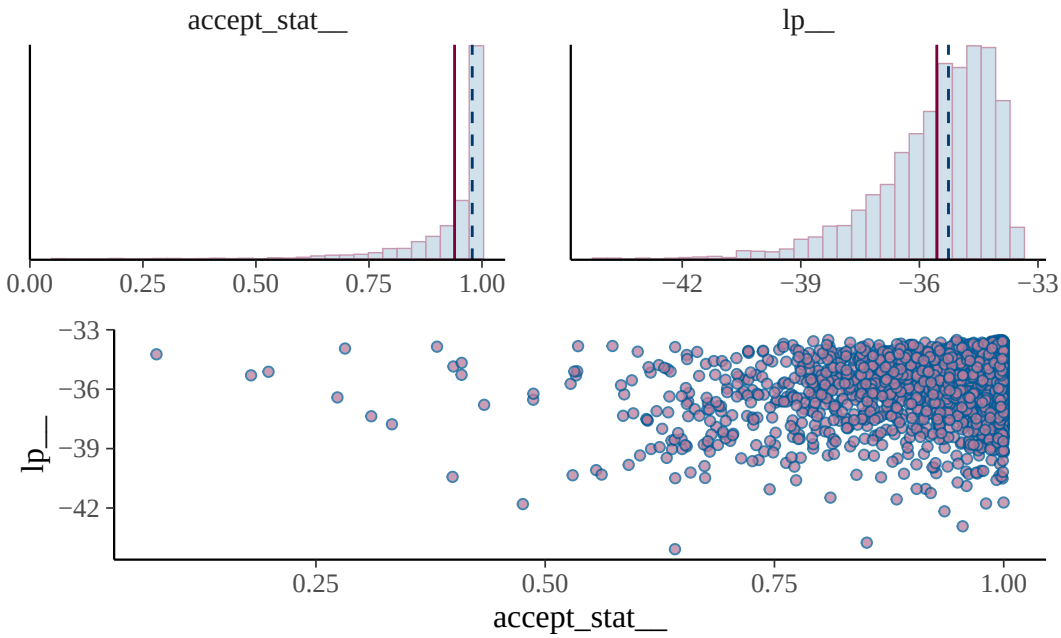
```r
mcmc_nuts_treedepth(nuts_fit, log_posterior(fit))
```



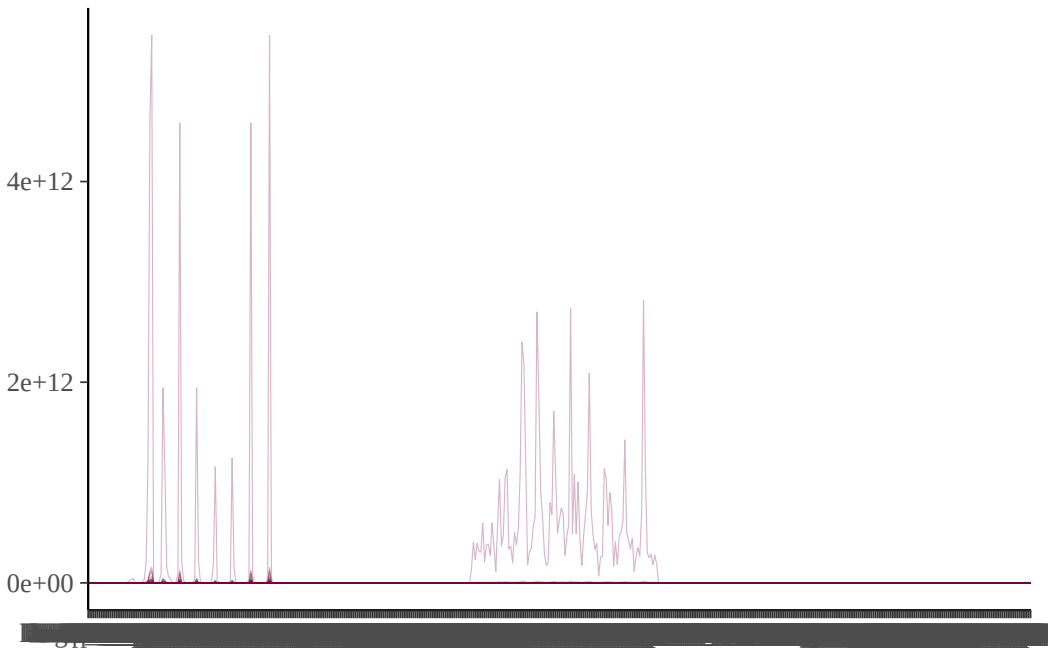


```r
mcmc_nuts_acceptance(nuts_fit,  log_posterior(fit))
```

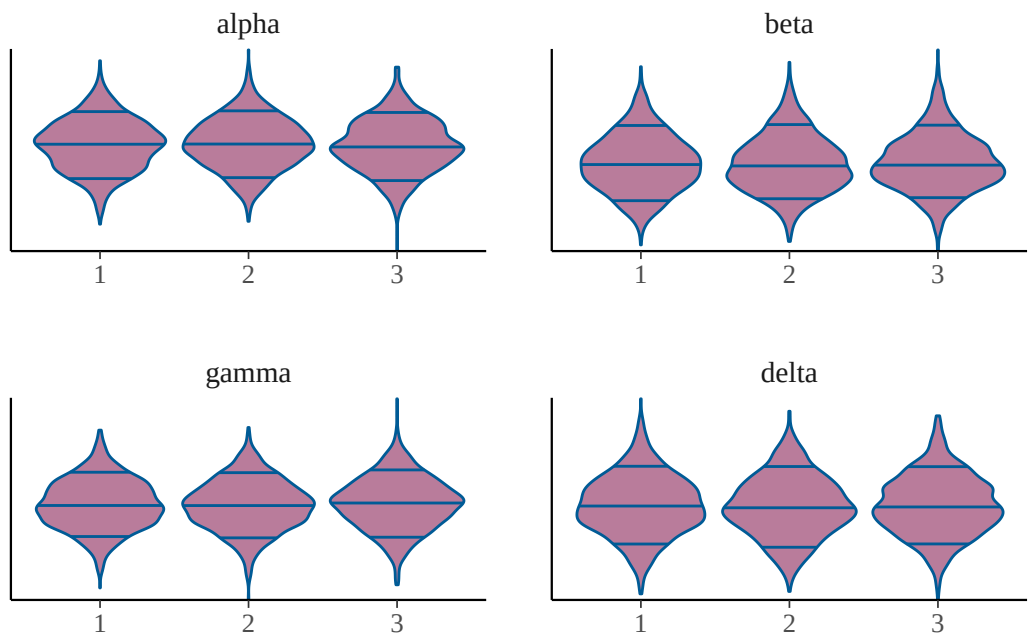
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



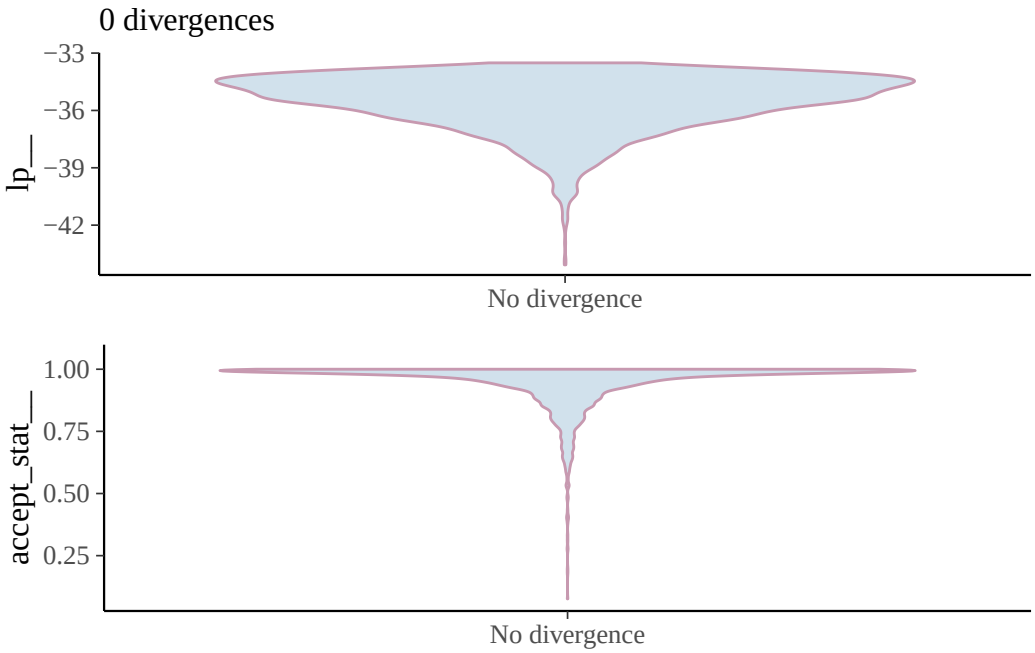

```r
mcmc_parcoord(fit$draws(), np=nuts_fit)
```



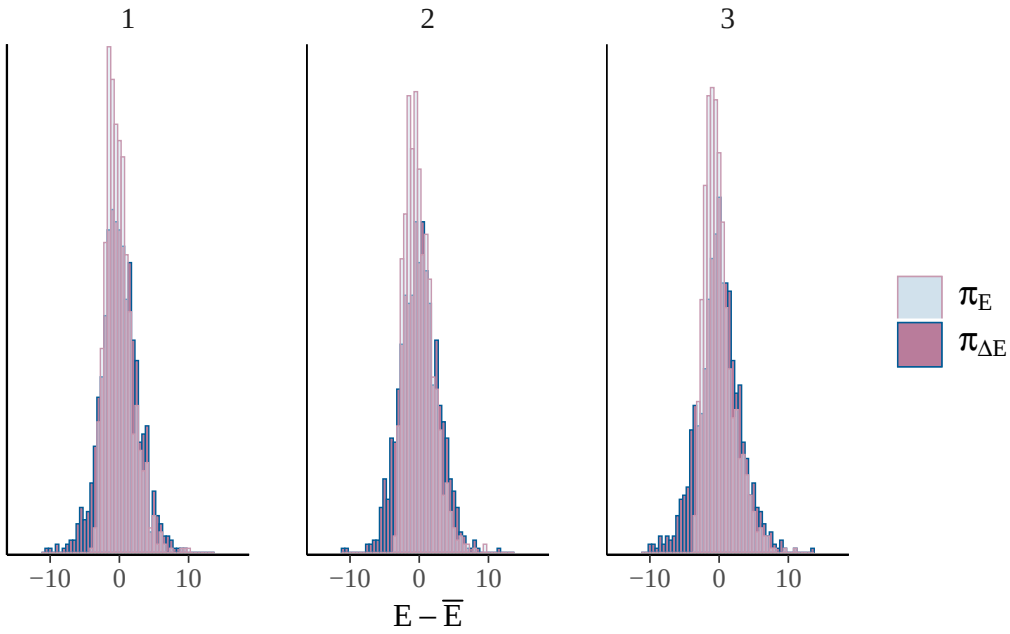

```
mcmc_violin(fit$draws(), pars = c('alpha', 'beta', 'gamma', 'delta'))
```



```r
mcmc_nuts_divergence(nuts_fit, log_posterior(fit))
```

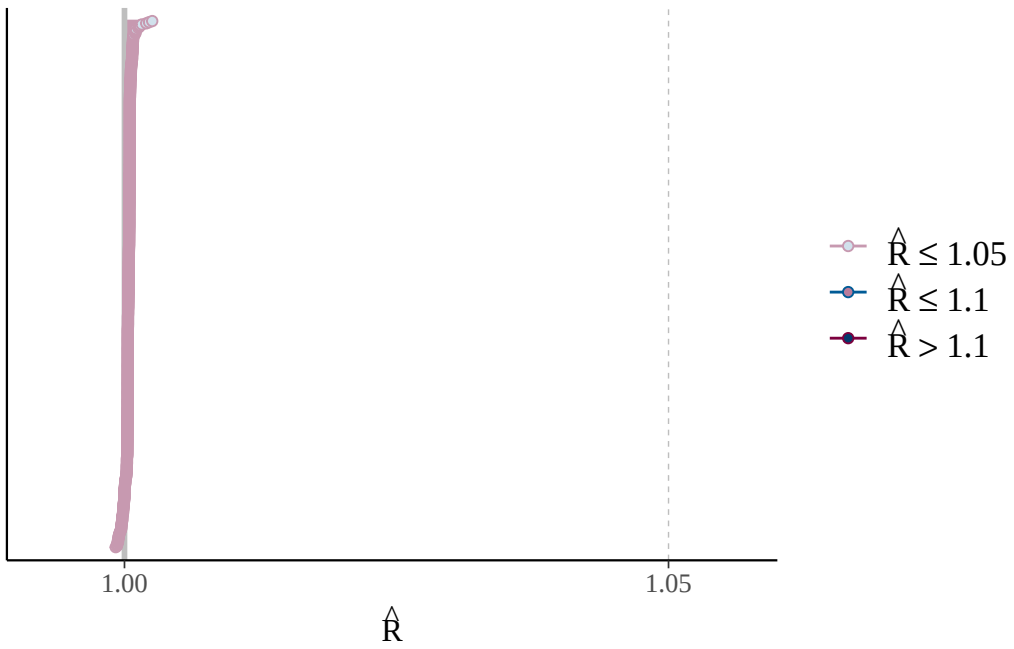



```r
mcmc_nuts_energy(nuts_fit, binwidth=1/2)
```



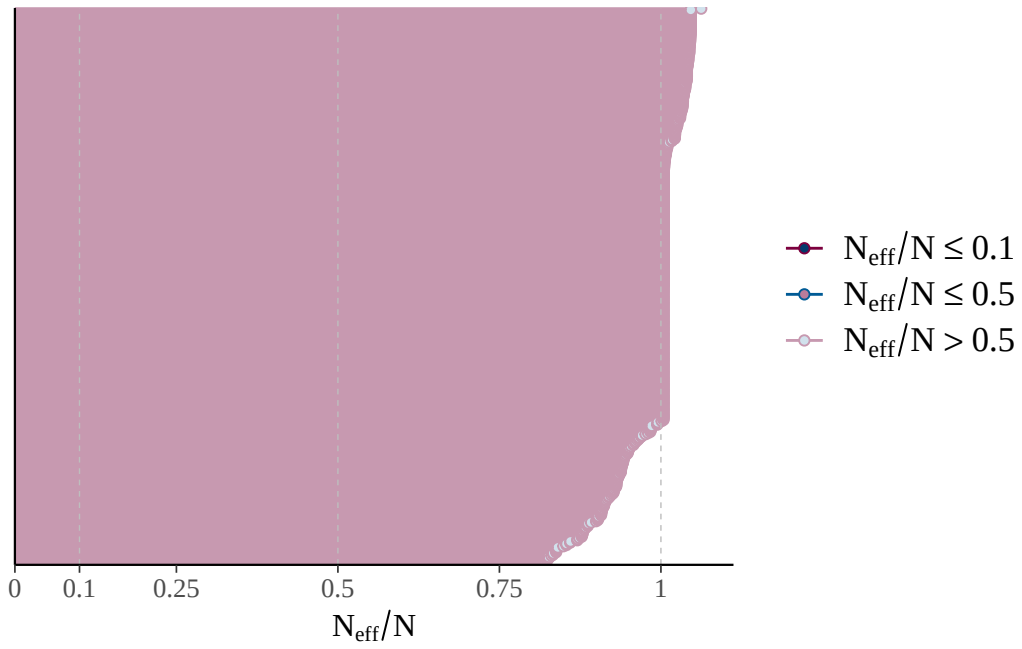

```
mcmc_rhat(rhat(fit))
```

```
Warning: Dropped 37 NAs from 'new_rhat(rhat)'.
```

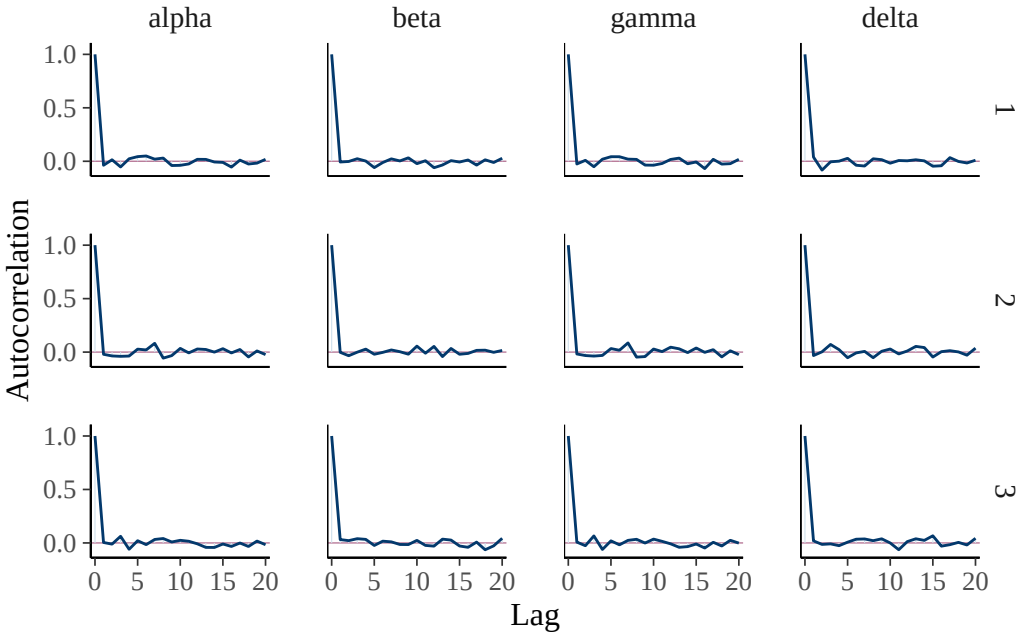



```
mcmc_neff(neff_ratio(fit))
```

```
Warning: Dropped 37 NAs from 'new_neff_ratio(ratio)'.
```




```
mcmc_acf(fit$draws(), pars=c('alpha', 'beta', 'gamma', 'delta'))
```



```r
# CMDSTAN functions for checking
fit$cmdstan_diagnose()
```

```
Processing csv files: C:/Users/msavery/AppData/Local/Temp/RtmpAPM5zE/thinned_poisson_process-

Checking sampler transitions treedepth.
Treedepth satisfactory for all transitions.

Checking sampler transitions for divergences.
No divergent transitions found.

Checking E-BFMI - sampler transitions HMC potential energy.
E-BFMI satisfactory.

Effective sample size satisfactory.

Split R-hat values satisfactory all parameters.

Processing complete, no problems detected.
```

```r
fit$cmdstan_summary()
```

```
Inference for Stan model: thinned_poisson_process_model
3 chains: each with iter=(1000,1000,1000); warmup=(0,0,0); thin=(10,10,10); 3000 iterations s

Warmup took (1.4, 0.98, 0.92) seconds, 3.3 seconds total
Sampling took (13, 13, 11) seconds, 37 seconds total

                         Mean     MCSE   StdDev        5%       50%      95%    N_Eff  N_Ef

lp__                 -3.6e+01  2.7e-02  1.4e+00  -3.8e+01  -3.5e+01 -3.4e+01  2.8e+03  7.5e-
accept_stat__            0.94  3.3e-03  9.3e-02      0.75      0.98     1.00  8.1e+02  2.2e-
stepsize__              0.018  1.6e-03  2.0e-03     0.016     0.018    0.021  1.5e+00  4.1e-
treedepth__               6.7  3.0e-02  1.5e+00       2.0       7.0      8.0  2.6e+03  6.9e-
n_leapfrog__              170  1.8e+00  8.5e+01       7.0       159      255  2.2e+03  5.8e-
divergent__              0.00      nan  0.0e+00      0.00      0.00     0.00      nan      n
energy__                   38  3.8e-02  2.0e+00        35        37       41  2.8e+03  7.7e-

alpha                -2.9e+00  1.3e-01  7.1e+00  -1.5e+01  -3.0e+00  8.8e+00  3.1e+03  8.4e-
beta                  2.7e+00  1.2e-02  6.2e-01   1.8e+00   2.7e+00  3.8e+00  2.8e+03  7.6e-
```

gamma	-2.8e+00	1.3e-01	7.2e+00	-1.4e+01	-2.9e+00	9.1e+00	3.1e+03	8.3e-
delta	4.3e-01	6.2e-03	3.4e-01	-1.2e-01	4.1e-01	1.0e+00	3.0e+03	8.0e-
lambda_rep[1]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.7e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[2]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.8e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[3]	1.6e+04	1.3e+04	7.3e+05	5.1e-10	9.3e-05	1.6e+01	3.0e+03	8.1e-
lambda_rep[4]	1.9e+04	1.6e+04	8.9e+05	6.4e-10	1.1e-04	1.8e+01	3.0e+03	8.1e-
lambda_rep[5]	3.4e+04	2.9e+04	1.6e+06	1.3e-09	2.2e-04	3.0e+01	3.0e+03	8.1e-
lambda_rep[6]	8.0e+04	6.9e+04	3.8e+06	3.5e-09	5.4e-04	7.7e+01	3.0e+03	8.1e-
lambda_rep[7]	1.9e+05	1.6e+05	8.9e+06	9.0e-09	1.3e-03	2.1e+02	3.0e+03	8.1e-
lambda_rep[8]	3.7e+05	3.3e+05	1.8e+07	2.0e-08	2.8e-03	4.2e+02	3.0e+03	8.1e-
lambda_rep[9]	5.8e+05	5.1e+05	2.8e+07	3.3e-08	4.5e-03	6.6e+02	3.0e+03	8.1e-
lambda_rep[10]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[11]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.8e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[12]	1.6e+04	1.3e+04	7.4e+05	5.2e-10	9.5e-05	1.6e+01	3.0e+03	8.1e-
lambda_rep[13]	2.2e+04	1.9e+04	1.0e+06	7.8e-10	1.3e-04	2.1e+01	3.0e+03	8.1e-
lambda_rep[14]	5.8e+04	5.0e+04	2.7e+06	2.4e-09	3.7e-04	5.3e+01	3.0e+03	8.1e-
lambda_rep[15]	2.1e+05	1.9e+05	1.0e+07	1.0e-08	1.5e-03	2.4e+02	3.0e+03	8.1e-
lambda_rep[16]	7.9e+05	7.0e+05	3.9e+07	4.9e-08	6.2e-03	9.6e+02	3.0e+03	8.1e-
lambda_rep[17]	2.6e+06	2.3e+06	1.3e+08	2.0e-07	2.4e-02	3.1e+03	3.0e+03	8.1e-
lambda_rep[18]	6.7e+06	6.1e+06	3.3e+08	5.2e-07	6.6e-02	8.3e+03	3.0e+03	8.1e-
lambda_rep[19]	1.3e+07	1.1e+07	6.3e+08	1.1e-06	1.4e-01	1.6e+04	3.0e+03	8.1e-
lambda_rep[20]	1.6e+07	1.4e+07	7.9e+08	1.4e-06	1.7e-01	2.0e+04	3.0e+03	8.1e-
lambda_rep[21]	1.6e+04	1.3e+04	7.3e+05	5.1e-10	9.3e-05	1.6e+01	3.0e+03	8.1e-
lambda_rep[22]	2.2e+04	1.9e+04	1.0e+06	7.8e-10	1.3e-04	2.1e+01	3.0e+03	8.1e-
lambda_rep[23]	7.2e+04	6.2e+04	3.4e+06	3.1e-09	4.7e-04	6.8e+01	3.0e+03	8.1e-
lambda_rep[24]	3.7e+05	3.3e+05	1.8e+07	2.0e-08	2.8e-03	4.2e+02	3.0e+03	8.1e-
lambda_rep[25]	2.2e+06	1.9e+06	1.1e+08	1.6e-07	1.9e-02	2.5e+03	3.0e+03	8.1e-
lambda_rep[26]	1.3e+07	1.1e+07	6.3e+08	1.1e-06	1.4e-01	1.6e+04	3.0e+03	8.1e-
lambda_rep[27]	7.6e+07	6.9e+07	3.8e+09	8.3e-06	9.3e-01	1.2e+05	3.0e+03	8.1e-
lambda_rep[28]	4.6e+08	4.1e+08	2.3e+10	5.6e-05	6.9e+00	8.5e+05	3.0e+03	8.1e-
lambda_rep[29]	1.7e+09	1.5e+09	8.4e+10	1.9e-04	2.8e+01	3.5e+06	3.0e+03	8.1e-
lambda_rep[30]	2.1e+09	1.8e+09	1.0e+11	2.3e-04	3.5e+01	4.1e+06	3.0e+03	8.1e-
lambda_rep[31]	1.9e+04	1.6e+04	8.9e+05	6.4e-10	1.1e-04	1.8e+01	3.0e+03	8.1e-
lambda_rep[32]	5.8e+04	5.0e+04	2.7e+06	2.4e-09	3.7e-04	5.3e+01	3.0e+03	8.1e-
lambda_rep[33]	3.7e+05	3.3e+05	1.8e+07	2.0e-08	2.8e-03	4.2e+02	3.0e+03	8.1e-
lambda_rep[34]	3.1e+06	2.8e+06	1.5e+08	2.5e-07	2.8e-02	3.7e+03	3.0e+03	8.1e-
lambda_rep[35]	3.3e+07	3.0e+07	1.6e+09	3.8e-06	3.8e-01	4.6e+04	3.0e+03	8.1e-
lambda_rep[36]	7.2e+08	6.5e+08	3.5e+10	8.8e-05	1.1e+01	1.4e+06	3.0e+03	8.1e-
lambda_rep[37]	4.3e+08	3.9e+08	2.1e+10	5.2e-05	6.3e+00	7.9e+05	3.0e+03	8.1e-
lambda_rep[38]	5.8e+07	5.3e+07	2.9e+09	6.6e-06	7.0e-01	8.9e+04	3.0e+03	8.1e-
lambda_rep[39]	2.2e+07	2.0e+07	1.1e+09	2.2e-06	2.4e-01	2.9e+04	3.0e+03	8.1e-
lambda_rep[40]	1.6e+07	1.4e+07	7.9e+08	1.4e-06	1.7e-01	2.0e+04	3.0e+03	8.1e-
lambda_rep[41]	3.4e+04	2.9e+04	1.6e+06	1.3e-09	2.2e-04	3.0e+01	3.0e+03	8.1e-

lambda_rep[42]	2.1e+05	1.9e+05	1.0e+07	1.0e-08	1.5e-03	2.4e+02	3.0e+03	8.1e-
lambda_rep[43]	2.2e+06	1.9e+06	1.1e+08	1.6e-07	1.9e-02	2.5e+03	3.0e+03	8.1e-
lambda_rep[44]	3.3e+07	3.0e+07	1.6e+09	3.8e-06	3.8e-01	4.6e+04	3.0e+03	8.1e-
lambda_rep[45]	1.7e+09	1.5e+09	8.4e+10	1.9e-04	2.8e+01	3.5e+06	3.0e+03	8.1e-
lambda_rep[46]	8.3e+07	7.5e+07	4.1e+09	9.3e-06	1.0e+00	1.3e+05	3.0e+03	8.1e-
lambda_rep[47]	8.7e+06	7.9e+06	4.3e+08	7.0e-07	8.9e-02	1.1e+04	3.0e+03	8.1e-
lambda_rep[48]	2.1e+06	1.9e+06	1.0e+08	1.6e-07	1.8e-02	2.4e+03	3.0e+03	8.1e-
lambda_rep[49]	9.0e+05	8.0e+05	4.4e+07	5.9e-08	7.1e-03	1.1e+03	3.0e+03	8.1e-
lambda_rep[50]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[51]	8.0e+04	6.9e+04	3.8e+06	3.5e-09	5.4e-04	7.7e+01	3.0e+03	8.1e-
lambda_rep[52]	7.9e+05	7.0e+05	3.9e+07	4.9e-08	6.2e-03	9.6e+02	3.0e+03	8.1e-
lambda_rep[53]	1.3e+07	1.1e+07	6.3e+08	1.1e-06	1.4e-01	1.6e+04	3.0e+03	8.1e-
lambda_rep[54]	7.2e+08	6.5e+08	3.5e+10	8.8e-05	1.1e+01	1.4e+06	3.0e+03	8.1e-
lambda_rep[55]	8.3e+07	7.5e+07	4.1e+09	9.3e-06	1.0e+00	1.3e+05	3.0e+03	8.1e-
lambda_rep[56]	4.9e+06	4.4e+06	2.4e+08	3.7e-07	4.7e-02	5.8e+03	3.0e+03	8.1e-
lambda_rep[57]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[58]	1.7e+05	1.5e+05	8.1e+06	8.2e-09	1.2e-03	1.8e+02	3.0e+03	8.1e-
lambda_rep[59]	7.5e+04	6.5e+04	3.6e+06	3.3e-09	5.0e-04	7.2e+01	3.0e+03	8.1e-
lambda_rep[60]	5.8e+04	5.0e+04	2.7e+06	2.4e-09	3.7e-04	5.3e+01	3.0e+03	8.1e-
lambda_rep[61]	1.9e+05	1.6e+05	8.9e+06	9.0e-09	1.3e-03	2.1e+02	3.0e+03	8.1e-
lambda_rep[62]	2.6e+06	2.3e+06	1.3e+08	2.0e-07	2.4e-02	3.1e+03	3.0e+03	8.1e-
lambda_rep[63]	7.6e+07	6.9e+07	3.8e+09	8.3e-06	9.3e-01	1.2e+05	3.0e+03	8.1e-
lambda_rep[64]	4.3e+08	3.9e+08	2.1e+10	5.2e-05	6.3e+00	7.9e+05	3.0e+03	8.1e-
lambda_rep[65]	8.7e+06	7.9e+06	4.3e+08	7.0e-07	8.9e-02	1.1e+04	3.0e+03	8.1e-
lambda_rep[66]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[67]	9.8e+04	8.5e+04	4.6e+06	4.5e-09	6.7e-04	9.8e+01	3.0e+03	8.1e-
lambda_rep[68]	3.0e+04	2.5e+04	1.4e+06	1.1e-09	1.8e-04	2.7e+01	3.0e+03	8.1e-
lambda_rep[69]	1.9e+04	1.6e+04	8.8e+05	6.3e-10	1.1e-04	1.8e+01	3.0e+03	8.1e-
lambda_rep[70]	1.7e+04	1.5e+04	8.1e+05	5.8e-10	1.0e-04	1.7e+01	3.0e+03	8.1e-
lambda_rep[71]	3.7e+05	3.3e+05	1.8e+07	2.0e-08	2.8e-03	4.2e+02	3.0e+03	8.1e-
lambda_rep[72]	6.7e+06	6.1e+06	3.3e+08	5.2e-07	6.6e-02	8.3e+03	3.0e+03	8.1e-
lambda_rep[73]	4.6e+08	4.1e+08	2.3e+10	5.6e-05	6.9e+00	8.5e+05	3.0e+03	8.1e-
lambda_rep[74]	5.8e+07	5.3e+07	2.9e+09	6.6e-06	7.0e-01	8.9e+04	3.0e+03	8.1e-
lambda_rep[75]	2.1e+06	1.9e+06	1.0e+08	1.6e-07	1.8e-02	2.4e+03	3.0e+03	8.1e-
lambda_rep[76]	1.7e+05	1.5e+05	8.1e+06	8.2e-09	1.2e-03	1.8e+02	3.0e+03	8.1e-
lambda_rep[77]	3.0e+04	2.5e+04	1.4e+06	1.1e-09	1.8e-04	2.7e+01	3.0e+03	8.1e-
lambda_rep[78]	1.6e+04	1.4e+04	7.6e+05	5.4e-10	9.7e-05	1.6e+01	3.0e+03	8.1e-
lambda_rep[79]	1.5e+04	1.3e+04	7.0e+05	5.0e-10	8.9e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[80]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.8e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[81]	5.8e+05	5.1e+05	2.8e+07	3.3e-08	4.5e-03	6.6e+02	3.0e+03	8.1e-
lambda_rep[82]	1.3e+07	1.1e+07	6.3e+08	1.1e-06	1.4e-01	1.6e+04	3.0e+03	8.1e-
lambda_rep[83]	1.7e+09	1.5e+09	8.4e+10	1.9e-04	2.8e+01	3.5e+06	3.0e+03	8.1e-
lambda_rep[84]	2.2e+07	2.0e+07	1.1e+09	2.2e-06	2.4e-01	2.9e+04	3.0e+03	8.1e-

lambda_rep[85]	9.0e+05	8.0e+05	4.4e+07	5.9e-08	7.1e-03	1.1e+03	3.0e+03	8.1e-
lambda_rep[86]	7.5e+04	6.5e+04	3.6e+06	3.3e-09	5.0e-04	7.2e+01	3.0e+03	8.1e-
lambda_rep[87]	1.9e+04	1.6e+04	8.8e+05	6.3e-10	1.1e-04	1.8e+01	3.0e+03	8.1e-
lambda_rep[88]	1.5e+04	1.3e+04	7.0e+05	5.0e-10	8.9e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[89]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.7e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[90]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.7e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[91]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[92]	1.6e+07	1.4e+07	7.9e+08	1.4e-06	1.7e-01	2.0e+04	3.0e+03	8.1e-
lambda_rep[93]	2.1e+09	1.8e+09	1.0e+11	2.3e-04	3.5e+01	4.1e+06	3.0e+03	8.1e-
lambda_rep[94]	1.6e+07	1.4e+07	7.9e+08	1.4e-06	1.7e-01	2.0e+04	3.0e+03	8.1e-
lambda_rep[95]	6.8e+05	6.0e+05	3.3e+07	4.0e-08	5.2e-03	8.1e+02	3.0e+03	8.1e-
lambda_rep[96]	5.8e+04	5.0e+04	2.7e+06	2.4e-09	3.7e-04	5.3e+01	3.0e+03	8.1e-
lambda_rep[97]	1.7e+04	1.5e+04	8.1e+05	5.8e-10	1.0e-04	1.7e+01	3.0e+03	8.1e-
lambda_rep[98]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.8e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[99]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.7e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep[100]	1.5e+04	1.3e+04	7.0e+05	4.9e-10	8.7e-05	1.5e+01	3.0e+03	8.1e-
lambda_rep_log[1]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[2]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[3]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[4]	-9.1e+00	1.3e-01	7.4e+00	-2.1e+01	-9.1e+00	2.9e+00	3.1e+03	8.5e-
lambda_rep_log[5]	-8.4e+00	1.3e-01	7.3e+00	-2.0e+01	-8.4e+00	3.4e+00	3.1e+03	8.4e-
lambda_rep_log[6]	-7.5e+00	1.3e-01	7.3e+00	-1.9e+01	-7.5e+00	4.3e+00	3.1e+03	8.4e-
lambda_rep_log[7]	-6.6e+00	1.3e-01	7.2e+00	-1.9e+01	-6.6e+00	5.3e+00	3.1e+03	8.4e-
lambda_rep_log[8]	-5.8e+00	1.3e-01	7.2e+00	-1.8e+01	-5.9e+00	6.0e+00	3.1e+03	8.4e-
lambda_rep_log[9]	-5.3e+00	1.3e-01	7.2e+00	-1.7e+01	-5.4e+00	6.5e+00	3.1e+03	8.4e-
lambda_rep_log[10]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[11]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[12]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.8e+00	3.1e+03	8.5e-
lambda_rep_log[13]	-8.9e+00	1.3e-01	7.4e+00	-2.1e+01	-8.9e+00	3.0e+00	3.1e+03	8.5e-
lambda_rep_log[14]	-7.9e+00	1.3e-01	7.3e+00	-2.0e+01	-7.9e+00	4.0e+00	3.1e+03	8.4e-
lambda_rep_log[15]	-6.4e+00	1.3e-01	7.2e+00	-1.8e+01	-6.5e+00	5.5e+00	3.1e+03	8.4e-
lambda_rep_log[16]	-5.0e+00	1.3e-01	7.2e+00	-1.7e+01	-5.1e+00	6.9e+00	3.1e+03	8.4e-
lambda_rep_log[17]	-3.7e+00	1.3e-01	7.2e+00	-1.5e+01	-3.7e+00	8.0e+00	3.1e+03	8.4e-
lambda_rep_log[18]	-2.7e+00	1.3e-01	7.1e+00	-1.4e+01	-2.7e+00	9.0e+00	3.1e+03	8.4e-
lambda_rep_log[19]	-2.0e+00	1.3e-01	7.1e+00	-1.4e+01	-2.0e+00	9.7e+00	3.1e+03	8.4e-
lambda_rep_log[20]	-1.7e+00	1.3e-01	7.1e+00	-1.3e+01	-1.7e+00	9.9e+00	3.1e+03	8.4e-
lambda_rep_log[21]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[22]	-8.9e+00	1.3e-01	7.4e+00	-2.1e+01	-8.9e+00	3.0e+00	3.1e+03	8.5e-
lambda_rep_log[23]	-7.6e+00	1.3e-01	7.3e+00	-2.0e+01	-7.7e+00	4.2e+00	3.1e+03	8.4e-
lambda_rep_log[24]	-5.8e+00	1.3e-01	7.2e+00	-1.8e+01	-5.9e+00	6.0e+00	3.1e+03	8.4e-
lambda_rep_log[25]	-3.9e+00	1.3e-01	7.2e+00	-1.6e+01	-4.0e+00	7.8e+00	3.1e+03	8.4e-
lambda_rep_log[26]	-2.0e+00	1.3e-01	7.1e+00	-1.4e+01	-2.0e+00	9.7e+00	3.1e+03	8.4e-
lambda_rep_log[27]	-5.8e-02	1.3e-01	7.1e+00	-1.2e+01	-7.1e-02	1.2e+01	3.1e+03	8.3e-

lambda_rep_log[28]	1.9e+00	1.3e-01	7.2e+00	-9.8e+00	1.9e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[29]	3.3e+00	1.3e-01	7.2e+00	-8.6e+00	3.3e+00	1.5e+01	3.1e+03	8.3e-
lambda_rep_log[30]	3.5e+00	1.3e-01	7.2e+00	-8.4e+00	3.5e+00	1.5e+01	3.1e+03	8.3e-
lambda_rep_log[31]	-9.1e+00	1.3e-01	7.4e+00	-2.1e+01	-9.1e+00	2.9e+00	3.1e+03	8.5e-
lambda_rep_log[32]	-7.9e+00	1.3e-01	7.3e+00	-2.0e+01	-7.9e+00	4.0e+00	3.1e+03	8.4e-
lambda_rep_log[33]	-5.8e+00	1.3e-01	7.2e+00	-1.8e+01	-5.9e+00	6.0e+00	3.1e+03	8.4e-
lambda_rep_log[34]	-3.5e+00	1.3e-01	7.2e+00	-1.5e+01	-3.6e+00	8.2e+00	3.1e+03	8.4e-
lambda_rep_log[35]	-9.6e-01	1.3e-01	7.1e+00	-1.2e+01	-9.7e-01	1.1e+01	3.1e+03	8.3e-
lambda_rep_log[36]	2.4e+00	1.3e-01	7.2e+00	-9.3e+00	2.4e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[37]	1.8e+00	1.3e-01	7.2e+00	-9.9e+00	1.8e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[38]	-3.5e-01	1.3e-01	7.1e+00	-1.2e+01	-3.6e-01	1.1e+01	3.1e+03	8.3e-
lambda_rep_log[39]	-1.4e+00	1.3e-01	7.1e+00	-1.3e+01	-1.4e+00	1.0e+01	3.1e+03	8.3e-
lambda_rep_log[40]	-1.7e+00	1.3e-01	7.1e+00	-1.3e+01	-1.7e+00	9.9e+00	3.1e+03	8.4e-
lambda_rep_log[41]	-8.4e+00	1.3e-01	7.3e+00	-2.0e+01	-8.4e+00	3.4e+00	3.1e+03	8.4e-
lambda_rep_log[42]	-6.4e+00	1.3e-01	7.2e+00	-1.8e+01	-6.5e+00	5.5e+00	3.1e+03	8.4e-
lambda_rep_log[43]	-3.9e+00	1.3e-01	7.2e+00	-1.6e+01	-4.0e+00	7.8e+00	3.1e+03	8.4e-
lambda_rep_log[44]	-9.6e-01	1.3e-01	7.1e+00	-1.2e+01	-9.7e-01	1.1e+01	3.1e+03	8.3e-
lambda_rep_log[45]	3.3e+00	1.3e-01	7.2e+00	-8.6e+00	3.3e+00	1.5e+01	3.1e+03	8.3e-
lambda_rep_log[46]	3.6e-02	1.3e-01	7.1e+00	-1.2e+01	4.0e-02	1.2e+01	3.1e+03	8.3e-
lambda_rep_log[47]	-2.4e+00	1.3e-01	7.1e+00	-1.4e+01	-2.4e+00	9.3e+00	3.1e+03	8.4e-
lambda_rep_log[48]	-4.0e+00	1.3e-01	7.2e+00	-1.6e+01	-4.0e+00	7.8e+00	3.1e+03	8.4e-
lambda_rep_log[49]	-4.9e+00	1.3e-01	7.2e+00	-1.7e+01	-4.9e+00	7.0e+00	3.1e+03	8.4e-
lambda_rep_log[50]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[51]	-7.5e+00	1.3e-01	7.3e+00	-1.9e+01	-7.5e+00	4.3e+00	3.1e+03	8.4e-
lambda_rep_log[52]	-5.0e+00	1.3e-01	7.2e+00	-1.7e+01	-5.1e+00	6.9e+00	3.1e+03	8.4e-
lambda_rep_log[53]	-2.0e+00	1.3e-01	7.1e+00	-1.4e+01	-2.0e+00	9.7e+00	3.1e+03	8.4e-
lambda_rep_log[54]	2.4e+00	1.3e-01	7.2e+00	-9.3e+00	2.4e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[55]	3.6e-02	1.3e-01	7.1e+00	-1.2e+01	4.0e-02	1.2e+01	3.1e+03	8.3e-
lambda_rep_log[56]	-3.0e+00	1.3e-01	7.1e+00	-1.5e+01	-3.1e+00	8.7e+00	3.1e+03	8.4e-
lambda_rep_log[57]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[58]	-6.7e+00	1.3e-01	7.2e+00	-1.9e+01	-6.7e+00	5.2e+00	3.1e+03	8.4e-
lambda_rep_log[59]	-7.6e+00	1.3e-01	7.3e+00	-2.0e+01	-7.6e+00	4.3e+00	3.1e+03	8.4e-
lambda_rep_log[60]	-7.9e+00	1.3e-01	7.3e+00	-2.0e+01	-7.9e+00	4.0e+00	3.1e+03	8.4e-
lambda_rep_log[61]	-6.6e+00	1.3e-01	7.2e+00	-1.9e+01	-6.6e+00	5.3e+00	3.1e+03	8.4e-
lambda_rep_log[62]	-3.7e+00	1.3e-01	7.2e+00	-1.5e+01	-3.7e+00	8.0e+00	3.1e+03	8.4e-
lambda_rep_log[63]	-5.8e-02	1.3e-01	7.1e+00	-1.2e+01	-7.1e-02	1.2e+01	3.1e+03	8.3e-
lambda_rep_log[64]	1.8e+00	1.3e-01	7.2e+00	-9.9e+00	1.8e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[65]	-2.4e+00	1.3e-01	7.1e+00	-1.4e+01	-2.4e+00	9.3e+00	3.1e+03	8.4e-
lambda_rep_log[66]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[67]	-7.3e+00	1.3e-01	7.3e+00	-1.9e+01	-7.3e+00	4.6e+00	3.1e+03	8.4e-
lambda_rep_log[68]	-8.6e+00	1.3e-01	7.3e+00	-2.1e+01	-8.6e+00	3.3e+00	3.1e+03	8.4e-
lambda_rep_log[69]	-9.1e+00	1.3e-01	7.4e+00	-2.1e+01	-9.1e+00	2.9e+00	3.1e+03	8.5e-
lambda_rep_log[70]	-9.2e+00	1.3e-01	7.4e+00	-2.1e+01	-9.2e+00	2.8e+00	3.1e+03	8.5e-

lambda_rep_log[71]	-5.8e+00	1.3e-01	7.2e+00	-1.8e+01	-5.9e+00	6.0e+00	3.1e+03	8.4e-
lambda_rep_log[72]	-2.7e+00	1.3e-01	7.1e+00	-1.4e+01	-2.7e+00	9.0e+00	3.1e+03	8.4e-
lambda_rep_log[73]	1.9e+00	1.3e-01	7.2e+00	-9.8e+00	1.9e+00	1.4e+01	3.1e+03	8.3e-
lambda_rep_log[74]	-3.5e-01	1.3e-01	7.1e+00	-1.2e+01	-3.6e-01	1.1e+01	3.1e+03	8.3e-
lambda_rep_log[75]	-4.0e+00	1.3e-01	7.2e+00	-1.6e+01	-4.0e+00	7.8e+00	3.1e+03	8.4e-
lambda_rep_log[76]	-6.7e+00	1.3e-01	7.2e+00	-1.9e+01	-6.7e+00	5.2e+00	3.1e+03	8.4e-
lambda_rep_log[77]	-8.6e+00	1.3e-01	7.3e+00	-2.1e+01	-8.6e+00	3.3e+00	3.1e+03	8.4e-
lambda_rep_log[78]	-9.2e+00	1.3e-01	7.4e+00	-2.1e+01	-9.2e+00	2.8e+00	3.1e+03	8.5e-
lambda_rep_log[79]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[80]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[81]	-5.3e+00	1.3e-01	7.2e+00	-1.7e+01	-5.4e+00	6.5e+00	3.1e+03	8.4e-
lambda_rep_log[82]	-2.0e+00	1.3e-01	7.1e+00	-1.4e+01	-2.0e+00	9.7e+00	3.1e+03	8.4e-
lambda_rep_log[83]	3.3e+00	1.3e-01	7.2e+00	-8.6e+00	3.3e+00	1.5e+01	3.1e+03	8.3e-
lambda_rep_log[84]	-1.4e+00	1.3e-01	7.1e+00	-1.3e+01	-1.4e+00	1.0e+01	3.1e+03	8.3e-
lambda_rep_log[85]	-4.9e+00	1.3e-01	7.2e+00	-1.7e+01	-4.9e+00	7.0e+00	3.1e+03	8.4e-
lambda_rep_log[86]	-7.6e+00	1.3e-01	7.3e+00	-2.0e+01	-7.6e+00	4.3e+00	3.1e+03	8.4e-
lambda_rep_log[87]	-9.1e+00	1.3e-01	7.4e+00	-2.1e+01	-9.1e+00	2.9e+00	3.1e+03	8.5e-
lambda_rep_log[88]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[89]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[90]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[91]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[92]	-1.7e+00	1.3e-01	7.1e+00	-1.3e+01	-1.7e+00	9.9e+00	3.1e+03	8.4e-
lambda_rep_log[93]	3.5e+00	1.3e-01	7.2e+00	-8.4e+00	3.5e+00	1.5e+01	3.1e+03	8.3e-
lambda_rep_log[94]	-1.7e+00	1.3e-01	7.1e+00	-1.3e+01	-1.7e+00	9.9e+00	3.1e+03	8.4e-
lambda_rep_log[95]	-5.2e+00	1.3e-01	7.2e+00	-1.7e+01	-5.3e+00	6.7e+00	3.1e+03	8.4e-
lambda_rep_log[96]	-7.9e+00	1.3e-01	7.3e+00	-2.0e+01	-7.9e+00	4.0e+00	3.1e+03	8.4e-
lambda_rep_log[97]	-9.2e+00	1.3e-01	7.4e+00	-2.1e+01	-9.2e+00	2.8e+00	3.1e+03	8.5e-
lambda_rep_log[98]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[99]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
lambda_rep_log[100]	-9.3e+00	1.3e-01	7.4e+00	-2.1e+01	-9.3e+00	2.7e+00	3.1e+03	8.5e-
b_rep[1]	5.1e+07	4.9e+07	2.7e+09	1.8e-07	2.2e-02	3.4e+03	3.0e+03	8.1e-
b_rep[2]	1.4e+08	1.4e+08	7.4e+09	4.5e-07	4.5e-02	6.7e+03	3.0e+03	8.1e-
b_rep[3]	7.9e+07	7.7e+07	4.2e+09	2.5e-07	3.1e-02	4.6e+03	3.0e+03	8.1e-
b_rep[4]	1.3e+08	1.3e+08	7.2e+09	4.4e-07	4.4e-02	6.4e+03	3.0e+03	8.1e-
b_rep[5]	1.1e+08	1.0e+08	5.7e+09	3.2e-07	3.7e-02	5.2e+03	3.0e+03	8.1e-
b_rep[6]	1.0e+08	1.0e+08	5.6e+09	3.1e-07	3.7e-02	5.2e+03	3.0e+03	8.1e-
b_rep[7]	2.0e+08	2.0e+08	1.1e+10	5.6e-07	5.6e-02	8.7e+03	3.0e+03	8.1e-
b_rep[8]	7.3e+07	7.1e+07	3.9e+09	2.5e-07	2.9e-02	4.5e+03	3.0e+03	8.1e-
b_rep[9]	1.3e+08	1.2e+08	6.8e+09	4.2e-07	4.2e-02	6.0e+03	3.0e+03	8.1e-
b_rep[10]	1.3e+08	1.3e+08	7.1e+09	4.3e-07	4.3e-02	6.3e+03	3.0e+03	8.1e-
b_rep[11]	9.4e+07	9.1e+07	5.0e+09	2.9e-07	3.4e-02	5.0e+03	3.0e+03	8.1e-
b_rep[12]	2.0e+08	2.0e+08	1.1e+10	5.6e-07	5.6e-02	8.7e+03	3.0e+03	8.1e-
b_rep[13]	1.2e+08	1.2e+08	6.6e+09	4.0e-07	4.1e-02	6.0e+03	3.0e+03	8.1e-

b_rep[14]	4.0e+07	3.8e+07	2.1e+09	1.3e-07	1.9e-02	2.8e+03	3.0e+03	8.1e-
b_rep[15]	2.0e+08	2.0e+08	1.1e+10	5.6e-07	5.6e-02	8.7e+03	3.0e+03	8.1e-
b_rep[16]	3.5e+08	3.4e+08	1.9e+10	7.4e-07	8.4e-02	1.1e+04	3.0e+03	8.1e-
b_rep[17]	1.3e+08	1.2e+08	6.7e+09	4.1e-07	4.1e-02	6.0e+03	3.0e+03	8.1e-
b_rep[18]	1.6e+08	1.6e+08	8.7e+09	5.0e-07	5.0e-02	7.3e+03	3.0e+03	8.1e-
b_rep[19]	3.5e+08	3.5e+08	1.9e+10	7.5e-07	8.5e-02	1.1e+04	3.0e+03	8.1e-
b_rep[20]	3.8e+08	3.8e+08	2.1e+10	7.8e-07	8.8e-02	1.1e+04	3.0e+03	8.1e-
b_rep[21]	1.1e+08	1.1e+08	6.2e+09	3.6e-07	3.9e-02	5.6e+03	3.0e+03	8.1e-
b_rep[22]	1.2e+08	1.2e+08	6.6e+09	4.0e-07	4.1e-02	6.0e+03	3.0e+03	8.1e-
b_rep[23]	7.0e+07	6.8e+07	3.7e+09	2.5e-07	2.8e-02	4.4e+03	3.0e+03	8.1e-
b_rep[24]	1.7e+08	1.7e+08	9.2e+09	5.1e-07	5.2e-02	7.7e+03	3.0e+03	8.1e-
b_rep[25]	1.3e+08	1.3e+08	7.0e+09	4.2e-07	4.2e-02	6.1e+03	3.0e+03	8.1e-
b_rep[26]	1.8e+08	1.8e+08	9.8e+09	5.2e-07	5.3e-02	8.3e+03	3.0e+03	8.1e-
b_rep[27]	3.6e+08	3.6e+08	2.0e+10	7.6e-07	8.6e-02	1.1e+04	3.0e+03	8.1e-
b_rep[28]	8.0e+08	8.0e+08	4.4e+10	1.3e-06	1.5e-01	2.0e+04	3.0e+03	8.1e-
b_rep[29]	7.4e+08	7.3e+08	4.0e+10	1.2e-06	1.4e-01	1.8e+04	3.0e+03	8.1e-
b_rep[30]	4.1e+08	4.0e+08	2.2e+10	8.3e-07	9.2e-02	1.2e+04	3.0e+03	8.1e-
b_rep[31]	6.2e+07	6.0e+07	3.3e+09	2.1e-07	2.6e-02	3.9e+03	3.0e+03	8.1e-
b_rep[32]	1.0e+08	1.0e+08	5.6e+09	3.1e-07	3.7e-02	5.2e+03	3.0e+03	8.1e-
b_rep[33]	1.2e+08	1.1e+08	6.3e+09	3.7e-07	4.0e-02	5.7e+03	3.0e+03	8.1e-
b_rep[34]	1.9e+08	1.9e+08	1.0e+10	5.3e-07	5.4e-02	8.6e+03	3.0e+03	8.1e-
b_rep[35]	2.2e+08	2.2e+08	1.2e+10	6.0e-07	6.1e-02	9.0e+03	3.0e+03	8.1e-
b_rep[36]	9.1e+08	9.0e+08	4.9e+10	1.4e-06	1.6e-01	2.2e+04	3.0e+03	8.1e-
b_rep[37]	6.3e+08	6.2e+08	3.4e+10	1.1e-06	1.3e-01	1.6e+04	3.0e+03	8.1e-
b_rep[38]	3.1e+08	3.1e+08	1.7e+10	7.0e-07	7.8e-02	1.0e+04	3.0e+03	8.1e-
b_rep[39]	2.2e+08	2.2e+08	1.2e+10	6.0e-07	6.1e-02	9.0e+03	3.0e+03	8.1e-
b_rep[40]	1.0e+08	9.7e+07	5.3e+09	3.0e-07	3.6e-02	5.1e+03	3.0e+03	8.1e-
b_rep[41]	6.0e+07	5.8e+07	3.2e+09	2.1e-07	2.5e-02	3.8e+03	3.0e+03	8.1e-
b_rep[42]	7.0e+07	6.8e+07	3.7e+09	2.5e-07	2.8e-02	4.4e+03	3.0e+03	8.1e-
b_rep[43]	2.7e+08	2.7e+08	1.5e+10	6.7e-07	7.1e-02	1.0e+04	3.0e+03	8.1e-
b_rep[44]	2.3e+08	2.3e+08	1.2e+10	6.1e-07	6.3e-02	8.9e+03	3.0e+03	8.1e-
b_rep[45]	5.7e+08	5.7e+08	3.1e+10	1.0e-06	1.2e-01	1.5e+04	3.0e+03	8.1e-
b_rep[46]	3.6e+08	3.6e+08	2.0e+10	7.6e-07	8.6e-02	1.1e+04	3.0e+03	8.1e-
b_rep[47]	1.7e+08	1.7e+08	9.0e+09	5.0e-07	5.1e-02	7.6e+03	3.0e+03	8.1e-
b_rep[48]	2.1e+08	2.1e+08	1.1e+10	5.9e-07	5.8e-02	9.2e+03	3.0e+03	8.1e-
b_rep[49]	2.5e+08	2.5e+08	1.4e+10	6.6e-07	6.7e-02	9.5e+03	3.0e+03	8.1e-
b_rep[50]	2.3e+08	2.3e+08	1.3e+10	6.1e-07	6.3e-02	8.8e+03	3.0e+03	8.1e-
b_rep[51]	9.4e+07	9.1e+07	5.0e+09	2.9e-07	3.4e-02	5.0e+03	3.0e+03	8.1e-
b_rep[52]	1.6e+08	1.5e+08	8.4e+09	4.9e-07	4.9e-02	7.0e+03	3.0e+03	8.1e-
b_rep[53]	2.0e+08	2.0e+08	1.1e+10	5.5e-07	5.6e-02	8.8e+03	3.0e+03	8.1e-
b_rep[54]	9.2e+08	9.1e+08	5.0e+10	1.4e-06	1.6e-01	2.3e+04	3.0e+03	8.1e-
b_rep[55]	1.7e+08	1.6e+08	8.9e+09	5.0e-07	5.1e-02	7.5e+03	3.0e+03	8.1e-
b_rep[56]	3.6e+08	3.6e+08	2.0e+10	7.6e-07	8.6e-02	1.1e+04	3.0e+03	8.1e-

b_rep[57]	1.7e+08	1.6e+08	9.0e+09	5.0e-07	5.1e-02	7.6e+03	3.0e+03	8.1e-
b_rep[58]	3.4e+08	3.3e+08	1.8e+10	7.3e-07	8.3e-02	1.0e+04	3.0e+03	8.1e-
b_rep[59]	1.5e+08	1.4e+08	7.8e+09	4.7e-07	4.6e-02	6.6e+03	3.0e+03	8.1e-
b_rep[60]	6.0e+07	5.8e+07	3.2e+09	2.1e-07	2.5e-02	3.8e+03	3.0e+03	8.1e-
b_rep[61]	1.6e+08	1.6e+08	8.8e+09	5.0e-07	5.1e-02	7.4e+03	3.0e+03	8.1e-
b_rep[62]	2.3e+08	2.3e+08	1.3e+10	6.1e-07	6.3e-02	8.9e+03	3.0e+03	8.1e-
b_rep[63]	3.1e+08	3.0e+08	1.7e+10	7.0e-07	7.8e-02	1.0e+04	3.0e+03	8.1e-
b_rep[64]	7.0e+08	7.0e+08	3.8e+10	1.2e-06	1.4e-01	1.8e+04	3.0e+03	8.1e-
b_rep[65]	2.5e+08	2.4e+08	1.3e+10	6.5e-07	6.6e-02	9.3e+03	3.0e+03	8.1e-
b_rep[66]	1.6e+08	1.6e+08	8.7e+09	5.0e-07	5.0e-02	7.3e+03	3.0e+03	8.1e-
b_rep[67]	1.1e+08	1.1e+08	6.1e+09	3.5e-07	3.9e-02	5.6e+03	3.0e+03	8.1e-
b_rep[68]	1.3e+08	1.3e+08	7.1e+09	4.3e-07	4.4e-02	6.4e+03	3.0e+03	8.1e-
b_rep[69]	2.4e+07	2.3e+07	1.3e+09	8.8e-08	1.3e-02	2.0e+03	3.0e+03	8.1e-
b_rep[70]	8.6e+07	8.4e+07	4.6e+09	2.7e-07	3.2e-02	4.8e+03	3.0e+03	8.1e-
b_rep[71]	9.0e+07	8.8e+07	4.8e+09	2.7e-07	3.3e-02	4.9e+03	3.0e+03	8.1e-
b_rep[72]	3.8e+08	3.8e+08	2.1e+10	7.8e-07	8.8e-02	1.1e+04	3.0e+03	8.1e-
b_rep[73]	3.5e+08	3.5e+08	1.9e+10	7.5e-07	8.5e-02	1.1e+04	3.0e+03	8.1e-
b_rep[74]	1.9e+08	1.9e+08	1.0e+10	5.4e-07	5.5e-02	8.7e+03	3.0e+03	8.1e-
b_rep[75]	3.0e+08	3.0e+08	1.6e+10	7.0e-07	7.8e-02	1.0e+04	3.0e+03	8.1e-
b_rep[76]	2.5e+08	2.4e+08	1.3e+10	6.5e-07	6.6e-02	9.4e+03	3.0e+03	8.1e-
b_rep[77]	5.7e+07	5.6e+07	3.0e+09	2.0e-07	2.5e-02	3.7e+03	3.0e+03	8.1e-
b_rep[78]	1.4e+08	1.4e+08	7.5e+09	4.5e-07	4.5e-02	6.7e+03	3.0e+03	8.1e-
b_rep[79]	6.3e+07	6.1e+07	3.3e+09	2.2e-07	2.6e-02	4.0e+03	3.0e+03	8.1e-
b_rep[80]	1.6e+08	1.5e+08	8.4e+09	4.9e-07	4.9e-02	7.1e+03	3.0e+03	8.1e-
b_rep[81]	1.7e+08	1.7e+08	9.3e+09	5.1e-07	5.2e-02	7.8e+03	3.0e+03	8.1e-
b_rep[82]	2.1e+08	2.0e+08	1.1e+10	5.8e-07	5.7e-02	9.1e+03	3.0e+03	8.1e-
b_rep[83]	4.8e+08	4.7e+08	2.6e+10	9.1e-07	1.0e-01	1.3e+04	3.0e+03	8.1e-
b_rep[84]	1.7e+08	1.7e+08	9.2e+09	5.1e-07	5.2e-02	7.8e+03	3.0e+03	8.1e-
b_rep[85]	1.4e+08	1.4e+08	7.7e+09	4.6e-07	4.6e-02	6.7e+03	3.0e+03	8.1e-
b_rep[86]	1.1e+08	1.1e+08	6.2e+09	3.6e-07	3.9e-02	5.6e+03	3.0e+03	8.1e-
b_rep[87]	1.5e+08	1.5e+08	8.1e+09	4.8e-07	4.7e-02	6.8e+03	3.0e+03	8.1e-
b_rep[88]	4.0e+07	3.9e+07	2.1e+09	1.3e-07	1.9e-02	2.9e+03	3.0e+03	8.1e-
b_rep[89]	9.0e+07	8.8e+07	4.8e+09	2.7e-07	3.3e-02	4.9e+03	3.0e+03	8.1e-
b_rep[90]	1.2e+08	1.2e+08	6.4e+09	3.8e-07	4.0e-02	5.8e+03	3.0e+03	8.1e-
b_rep[91]	9.1e+07	8.9e+07	4.9e+09	2.8e-07	3.3e-02	4.9e+03	3.0e+03	8.1e-
b_rep[92]	2.4e+08	2.4e+08	1.3e+10	6.5e-07	6.4e-02	9.0e+03	3.0e+03	8.1e-
b_rep[93]	9.5e+08	9.4e+08	5.1e+10	1.4e-06	1.6e-01	2.4e+04	3.0e+03	8.1e-
b_rep[94]	3.8e+08	3.7e+08	2.1e+10	7.7e-07	8.8e-02	1.1e+04	3.0e+03	8.1e-
b_rep[95]	1.0e+08	1.0e+08	5.5e+09	3.1e-07	3.6e-02	5.2e+03	3.0e+03	8.1e-
b_rep[96]	8.6e+07	8.3e+07	4.6e+09	2.7e-07	3.2e-02	4.8e+03	3.0e+03	8.1e-
b_rep[97]	9.7e+07	9.5e+07	5.2e+09	2.9e-07	3.5e-02	5.0e+03	3.0e+03	8.1e-
b_rep[98]	6.3e+07	6.1e+07	3.3e+09	2.2e-07	2.6e-02	4.0e+03	3.0e+03	8.1e-
b_rep[99]	9.3e+07	9.1e+07	5.0e+09	2.8e-07	3.4e-02	5.0e+03	3.0e+03	8.1e-

b_rep[100]	6.5e+07	6.3e+07	3.4e+09	2.2e-07	2.7e-02	4.0e+03	3.0e+03	8.1e-
lambda_bias[1]	2.7e-05	1.9e-06	9.9e-05	1.4e-08	2.7e-06	1.3e-04	2.7e+03	7.3e-
lambda_bias[2]	5.0e-05	3.8e-06	1.9e-04	3.4e-08	5.2e-06	2.3e-04	2.4e+03	6.6e-
lambda_bias[3]	3.5e-05	2.5e-06	1.3e-04	2.3e-08	3.7e-06	1.7e-04	2.5e+03	6.9e-
lambda_bias[4]	5.8e-05	4.1e-06	2.1e-04	4.8e-08	6.6e-06	2.7e-04	2.5e+03	6.7e-
lambda_bias[5]	7.5e-05	4.7e-06	2.4e-04	9.3e-08	1.1e-05	3.6e-04	2.5e+03	6.9e-
lambda_bias[6]	1.4e-04	7.6e-06	3.9e-04	3.4e-07	2.6e-05	6.7e-04	2.6e+03	7.1e-
lambda_bias[7]	4.2e-04	2.0e-05	1.0e-03	2.0e-06	1.0e-04	1.9e-03	2.6e+03	7.1e-
lambda_bias[8]	4.0e-04	1.7e-05	9.0e-04	2.2e-06	1.0e-04	1.8e-03	2.8e+03	7.6e-
lambda_bias[9]	7.6e-04	2.9e-05	1.5e-03	7.6e-06	2.5e-04	3.4e-03	2.7e+03	7.3e-
lambda_bias[10]	8.8e-04	3.2e-05	1.7e-03	9.9e-06	3.0e-04	3.9e-03	2.7e+03	7.3e-
lambda_bias[11]	3.8e-05	2.8e-06	1.4e-04	2.5e-08	4.0e-06	1.8e-04	2.5e+03	6.8e-
lambda_bias[12]	7.1e-05	5.6e-06	2.8e-04	5.1e-08	7.3e-06	3.1e-04	2.5e+03	6.7e-
lambda_bias[13]	6.1e-05	4.2e-06	2.1e-04	5.6e-08	7.2e-06	2.8e-04	2.5e+03	6.7e-
lambda_bias[14]	7.0e-05	4.1e-06	2.2e-04	7.7e-08	9.2e-06	3.2e-04	2.9e+03	7.8e-
lambda_bias[15]	4.6e-04	2.1e-05	1.1e-03	2.5e-06	1.2e-04	2.1e-03	2.6e+03	7.1e-
lambda_bias[16]	1.9e-03	7.3e-05	3.8e-03	2.7e-05	7.0e-04	8.0e-03	2.8e+03	7.5e-
lambda_bias[17]	2.7e-03	7.8e-05	4.1e-03	6.2e-05	1.2e-03	1.0e-02	2.8e+03	7.5e-
lambda_bias[18]	7.0e-03	1.7e-04	8.9e-03	3.3e-04	3.9e-03	2.4e-02	2.8e+03	7.5e-
lambda_bias[19]	2.0e-02	3.9e-04	2.1e-02	1.8e-03	1.4e-02	6.1e-02	2.8e+03	7.6e-
lambda_bias[20]	2.6e-02	4.7e-04	2.5e-02	2.6e-03	1.8e-02	7.5e-02	2.8e+03	7.6e-
lambda_bias[21]	4.5e-05	3.3e-06	1.6e-04	3.1e-08	4.7e-06	2.1e-04	2.5e+03	6.7e-
lambda_bias[22]	6.1e-05	4.2e-06	2.1e-04	5.6e-08	7.3e-06	2.8e-04	2.5e+03	6.7e-
lambda_bias[23]	1.0e-04	5.7e-06	3.0e-04	2.0e-07	1.8e-05	5.0e-04	2.7e+03	7.3e-
lambda_bias[24]	6.4e-04	2.6e-05	1.3e-03	5.0e-06	1.9e-04	2.9e-03	2.7e+03	7.2e-
lambda_bias[25]	2.3e-03	7.0e-05	3.7e-03	5.1e-05	1.0e-03	9.2e-03	2.8e+03	7.5e-
lambda_bias[26]	1.3e-02	2.8e-04	1.5e-02	9.3e-04	8.3e-03	4.2e-02	2.8e+03	7.5e-
lambda_bias[27]	1.1e-01	1.3e-03	6.7e-02	2.5e-02	9.2e-02	2.3e-01	2.8e+03	7.6e-
lambda_bias[28]	1.1e+00	7.1e-03	3.8e-01	5.7e-01	1.1e+00	1.8e+00	2.8e+03	7.7e-
lambda_bias[29]	4.0e+00	1.7e-02	8.8e-01	2.7e+00	4.0e+00	5.6e+00	2.8e+03	7.5e-
lambda_bias[30]	3.3e+00	2.3e-02	1.3e+00	1.6e+00	3.1e+00	5.7e+00	3.0e+03	8.0e-
lambda_bias[31]	3.6e-05	2.4e-06	1.2e-04	2.5e-08	3.9e-06	1.7e-04	2.6e+03	7.1e-
lambda_bias[32]	1.1e-04	6.3e-06	3.2e-04	2.0e-07	1.8e-05	5.3e-04	2.6e+03	7.0e-
lambda_bias[33]	5.1e-04	2.1e-05	1.1e-03	3.7e-06	1.5e-04	2.3e-03	2.7e+03	7.3e-
lambda_bias[34]	3.9e-03	1.0e-04	5.5e-03	1.3e-04	1.9e-03	1.5e-02	2.7e+03	7.4e-
lambda_bias[35]	3.6e-02	6.0e-04	3.2e-02	4.6e-03	2.7e-02	9.9e-02	2.8e+03	7.5e-
lambda_bias[36]	1.9e+00	1.1e-02	6.1e-01	1.1e+00	1.9e+00	3.0e+00	2.8e+03	7.6e-
lambda_bias[37]	8.4e-01	4.7e-03	2.5e-01	4.5e-01	8.3e-01	1.3e+00	2.9e+03	7.9e-
lambda_bias[38]	7.4e-02	9.9e-04	5.2e-02	1.4e-02	6.2e-02	1.8e-01	2.8e+03	7.6e-
lambda_bias[39]	2.4e-02	4.4e-04	2.3e-02	2.5e-03	1.7e-02	7.0e-02	2.8e+03	7.5e-
lambda_bias[40]	1.3e-02	3.2e-04	1.7e-02	6.0e-04	6.8e-03	4.4e-02	2.8e+03	7.7e-
lambda_bias[41]	5.5e-05	3.4e-06	1.8e-04	5.7e-08	7.3e-06	2.6e-04	2.7e+03	7.3e-
lambda_bias[42]	2.5e-04	1.2e-05	6.1e-04	9.4e-07	5.5e-05	1.1e-03	2.8e+03	7.5e-

lambda_bias[43]	3.6e-03	1.0e-04	5.5e-03	9.9e-05	1.7e-03	1.4e-02	2.7e+03	7.4e-
lambda_bias[44]	3.6e-02	6.0e-04	3.2e-02	4.7e-03	2.7e-02	9.9e-02	2.8e+03	7.5e-
lambda_bias[45]	3.4e+00	1.4e-02	7.9e-01	2.2e+00	3.3e+00	4.8e+00	3.0e+03	8.1e-
lambda_bias[46]	1.1e-01	1.3e-03	7.1e-02	2.8e-02	1.0e-01	2.5e-01	2.8e+03	7.6e-
lambda_bias[47]	9.1e-03	2.1e-04	1.1e-02	5.0e-04	5.3e-03	3.0e-02	2.8e+03	7.5e-
lambda_bias[48]	3.0e-03	8.6e-05	4.5e-03	7.7e-05	1.4e-03	1.2e-02	2.7e+03	7.4e-
lambda_bias[49]	1.7e-03	5.8e-05	3.0e-03	2.5e-05	6.5e-04	7.0e-03	2.7e+03	7.3e-
lambda_bias[50]	1.3e-03	4.7e-05	2.4e-03	1.6e-05	4.6e-04	5.3e-03	2.7e+03	7.3e-
lambda_bias[51]	1.3e-04	7.1e-06	3.7e-04	3.1e-07	2.4e-05	6.3e-04	2.6e+03	7.2e-
lambda_bias[52]	1.1e-03	3.9e-05	2.0e-03	1.4e-05	4.0e-04	4.7e-03	2.7e+03	7.3e-
lambda_bias[53]	1.4e-02	2.9e-04	1.5e-02	1.0e-03	8.9e-03	4.3e-02	2.8e+03	7.5e-
lambda_bias[54]	1.9e+00	1.2e-02	6.3e-01	1.1e+00	1.9e+00	3.1e+00	2.8e+03	7.6e-
lambda_bias[55]	7.6e-02	1.3e-03	6.7e-02	1.1e-02	5.7e-02	2.1e-01	2.8e+03	7.6e-
lambda_bias[56]	8.9e-03	2.2e-04	1.2e-02	4.2e-04	5.0e-03	3.1e-02	2.8e+03	7.6e-
lambda_bias[57]	1.0e-03	3.7e-05	1.9e-03	1.2e-05	3.5e-04	4.4e-03	2.7e+03	7.3e-
lambda_bias[58]	5.9e-04	3.1e-05	1.6e-03	2.5e-06	1.3e-04	2.7e-03	2.7e+03	7.3e-
lambda_bias[59]	1.7e-04	9.2e-06	4.7e-04	4.0e-07	3.0e-05	7.8e-04	2.6e+03	6.9e-
lambda_bias[60]	8.2e-05	4.7e-06	2.5e-04	1.3e-07	1.3e-05	3.9e-04	2.8e+03	7.4e-
lambda_bias[61]	3.6e-04	1.7e-05	8.6e-04	1.7e-06	8.8e-05	1.7e-03	2.6e+03	7.1e-
lambda_bias[62]	3.8e-03	1.0e-04	5.5e-03	1.2e-04	1.9e-03	1.4e-02	2.7e+03	7.4e-
lambda_bias[63]	9.5e-02	1.2e-03	6.3e-02	2.1e-02	8.1e-02	2.2e-01	2.8e+03	7.6e-
lambda_bias[64]	9.1e-01	5.5e-03	2.9e-01	4.8e-01	8.9e-01	1.4e+00	2.9e+03	7.8e-
lambda_bias[65]	1.1e-02	2.4e-04	1.3e-02	7.5e-04	7.0e-03	3.6e-02	2.8e+03	7.5e-
lambda_bias[66]	9.9e-04	3.6e-05	1.9e-03	1.2e-05	3.5e-04	4.3e-03	2.7e+03	7.3e-
lambda_bias[67]	1.7e-04	9.0e-06	4.6e-04	4.8e-07	3.4e-05	8.3e-04	2.6e+03	7.1e-
lambda_bias[68]	7.9e-05	5.2e-06	2.6e-04	8.9e-08	1.0e-05	3.7e-04	2.5e+03	6.8e-
lambda_bias[69]	2.4e-05	1.7e-06	9.3e-05	9.1e-09	1.9e-06	1.1e-04	3.0e+03	8.0e-
lambda_bias[70]	4.0e-05	2.8e-06	1.4e-04	2.8e-08	4.5e-06	1.9e-04	2.5e+03	6.8e-
lambda_bias[71]	4.4e-04	1.8e-05	9.6e-04	2.8e-06	1.2e-04	2.0e-03	2.8e+03	7.5e-
lambda_bias[72]	1.2e-02	2.8e-04	1.5e-02	7.0e-04	7.3e-03	4.1e-02	2.8e+03	7.6e-
lambda_bias[73]	6.2e-01	4.6e-03	2.5e-01	2.7e-01	5.9e-01	1.1e+00	3.0e+03	8.1e-
lambda_bias[74]	5.8e-02	9.2e-04	4.9e-02	8.2e-03	4.4e-02	1.5e-01	2.8e+03	7.6e-
lambda_bias[75]	3.8e-03	1.1e-04	5.9e-03	1.0e-04	1.8e-03	1.4e-02	2.8e+03	7.5e-
lambda_bias[76]	4.5e-04	2.3e-05	1.2e-03	2.1e-06	1.0e-04	2.0e-03	2.6e+03	7.1e-
lambda_bias[77]	4.8e-05	3.0e-06	1.6e-04	4.4e-08	6.0e-06	2.3e-04	2.7e+03	7.3e-
lambda_bias[78]	5.4e-05	3.9e-06	2.0e-04	3.9e-08	5.7e-06	2.5e-04	2.5e+03	6.6e-
lambda_bias[79]	3.0e-05	2.1e-06	1.1e-04	1.9e-08	3.1e-06	1.5e-04	2.6e+03	7.1e-
lambda_bias[80]	5.6e-05	4.2e-06	2.1e-04	3.8e-08	5.8e-06	2.5e-04	2.4e+03	6.6e-
lambda_bias[81]	9.1e-04	3.4e-05	1.8e-03	9.8e-06	3.1e-04	4.0e-03	2.7e+03	7.2e-
lambda_bias[82]	1.4e-02	2.9e-04	1.5e-02	1.1e-03	9.3e-03	4.4e-02	2.8e+03	7.5e-
lambda_bias[83]	3.0e+00	1.6e-02	8.7e-01	1.8e+00	2.9e+00	4.6e+00	3.0e+03	8.2e-
lambda_bias[84]	2.1e-02	4.1e-04	2.2e-02	1.8e-03	1.4e-02	6.4e-02	2.8e+03	7.5e-
lambda_bias[85]	1.2e-03	4.0e-05	2.1e-03	1.6e-05	4.3e-04	5.0e-03	2.7e+03	7.3e-

lambda_bias[86]	1.4e-04	7.8e-06	4.0e-04	3.3e-07	2.6e-05	6.8e-04	2.6e+03	7.0e-
lambda_bias[87]	6.3e-05	4.6e-06	2.3e-04	5.2e-08	7.1e-06	2.9e-04	2.5e+03	6.7e-
lambda_bias[88]	2.5e-05	1.7e-06	9.1e-05	1.2e-08	2.3e-06	1.1e-04	2.8e+03	7.5e-
lambda_bias[89]	3.7e-05	2.7e-06	1.3e-04	2.4e-08	3.9e-06	1.8e-04	2.5e+03	6.8e-
lambda_bias[90]	4.5e-05	3.3e-06	1.6e-04	3.0e-08	4.6e-06	2.1e-04	2.5e+03	6.6e-
lambda_bias[91]	7.3e-04	2.8e-05	1.5e-03	6.7e-06	2.3e-04	3.1e-03	2.8e+03	7.5e-
lambda_bias[92]	1.9e-02	3.6e-04	1.9e-02	1.7e-03	1.3e-02	5.7e-02	2.8e+03	7.5e-
lambda_bias[93]	6.0e+00	3.4e-02	1.8e+00	3.4e+00	5.8e+00	9.2e+00	2.7e+03	7.4e-
lambda_bias[94]	2.6e-02	4.7e-04	2.5e-02	2.5e-03	1.8e-02	7.5e-02	2.8e+03	7.6e-
lambda_bias[95]	7.7e-04	2.9e-05	1.5e-03	7.7e-06	2.5e-04	3.3e-03	2.8e+03	7.5e-
lambda_bias[96]	9.8e-05	5.6e-06	2.9e-04	1.8e-07	1.6e-05	4.7e-04	2.6e+03	7.1e-
lambda_bias[97]	4.3e-05	3.1e-06	1.5e-04	3.2e-08	4.8e-06	2.1e-04	2.5e+03	6.8e-
lambda_bias[98]	3.0e-05	2.1e-06	1.1e-04	1.8e-08	3.0e-06	1.4e-04	2.6e+03	7.1e-
lambda_bias[99]	3.8e-05	2.7e-06	1.4e-04	2.5e-08	4.0e-06	1.8e-04	2.5e+03	6.8e-
lambda_bias[100]	3.1e-05	2.2e-06	1.1e-04	1.9e-08	3.1e-06	1.5e-04	2.6e+03	7.0e-
y_rep[1]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[2]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[3]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[4]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[5]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[6]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[7]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[8]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[9]	6.7e-04	nan	2.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[10]	1.3e-03	nan	3.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[11]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[12]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[13]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[14]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[15]	6.7e-04	nan	2.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[16]	3.0e-03	9.9e-04	5.5e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.2e-
y_rep[17]	2.0e-03	nan	4.5e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[18]	7.0e-03	1.5e-03	8.3e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.2e-
y_rep[19]	1.7e-02	2.4e-03	1.3e-01	0.0e+00	0.0e+00	0.0e+00	2.8e+03	7.7e-
y_rep[20]	2.1e-02	2.7e-03	1.5e-01	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.1e-
y_rep[21]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[22]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[23]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[24]	6.7e-04	nan	2.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[25]	2.7e-03	nan	5.2e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[26]	1.4e-02	2.2e-03	1.2e-01	0.0e+00	0.0e+00	0.0e+00	2.9e+03	7.9e-
y_rep[27]	1.1e-01	6.1e-03	3.4e-01	0.0e+00	0.0e+00	1.0e+00	3.1e+03	8.3e-
y_rep[28]	1.1e+00	2.1e-02	1.1e+00	0.0e+00	1.0e+00	3.0e+00	3.0e+03	8.0e-

y_rep[29]	4.1e+00	4.1e-02	2.2e+00	1.0e+00	4.0e+00	8.0e+00	3.0e+03	8.1e-
y_rep[30]	3.3e+00	4.1e-02	2.2e+00	0.0e+00	3.0e+00	7.0e+00	2.9e+03	7.8e-
y_rep[31]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[32]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[33]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[34]	3.7e-03	1.2e-03	6.6e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.2e-
y_rep[35]	3.3e-02	3.3e-03	1.8e-01	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.1e-
y_rep[36]	1.9e+00	3.0e-02	1.6e+00	0.0e+00	2.0e+00	5.0e+00	2.6e+03	7.1e-
y_rep[37]	8.5e-01	1.7e-02	9.4e-01	0.0e+00	1.0e+00	3.0e+00	3.1e+03	8.3e-
y_rep[38]	7.2e-02	5.2e-03	2.8e-01	0.0e+00	0.0e+00	1.0e+00	2.9e+03	7.7e-
y_rep[39]	2.4e-02	2.9e-03	1.6e-01	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.0e-
y_rep[40]	1.4e-02	2.2e-03	1.2e-01	0.0e+00	0.0e+00	0.0e+00	2.9e+03	8.0e-
y_rep[41]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[42]	6.7e-04	nan	2.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[43]	3.7e-03	1.1e-03	6.0e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.2e-
y_rep[44]	3.9e-02	3.7e-03	2.0e-01	0.0e+00	0.0e+00	0.0e+00	2.9e+03	7.8e-
y_rep[45]	3.4e+00	3.5e-02	2.0e+00	1.0e+00	3.0e+00	7.0e+00	3.2e+03	8.6e-
y_rep[46]	1.1e-01	6.4e-03	3.4e-01	0.0e+00	0.0e+00	1.0e+00	2.9e+03	7.9e-
y_rep[47]	1.0e-02	1.9e-03	1.0e-01	0.0e+00	0.0e+00	0.0e+00	2.8e+03	7.6e-
y_rep[48]	1.0e-03	nan	3.2e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[49]	1.3e-03	nan	3.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[50]	1.0e-03	nan	3.2e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[51]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[52]	2.3e-03	8.8e-04	4.8e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.1e-
y_rep[53]	1.4e-02	2.3e-03	1.2e-01	0.0e+00	0.0e+00	0.0e+00	2.7e+03	7.4e-
y_rep[54]	2.0e+00	2.8e-02	1.5e+00	0.0e+00	2.0e+00	5.0e+00	3.0e+03	8.0e-
y_rep[55]	7.6e-02	5.4e-03	2.8e-01	0.0e+00	0.0e+00	1.0e+00	2.8e+03	7.5e-
y_rep[56]	9.3e-03	1.8e-03	1.0e-01	0.0e+00	0.0e+00	0.0e+00	3.1e+03	8.3e-
y_rep[57]	2.0e-03	8.1e-04	4.5e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.1e-
y_rep[58]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[59]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[60]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[61]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[62]	4.7e-03	1.2e-03	6.8e-02	0.0e+00	0.0e+00	0.0e+00	3.0e+03	8.2e-
y_rep[63]	9.4e-02	5.7e-03	3.1e-01	0.0e+00	0.0e+00	1.0e+00	2.9e+03	7.8e-
y_rep[64]	9.4e-01	1.9e-02	1.0e+00	0.0e+00	1.0e+00	3.0e+00	2.7e+03	7.4e-
y_rep[65]	1.4e-02	2.2e-03	1.2e-01	0.0e+00	0.0e+00	0.0e+00	3.1e+03	8.3e-
y_rep[66]	6.7e-04	nan	2.6e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[67]	3.3e-04	nan	1.8e-02	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[68]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[69]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[70]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n
y_rep[71]	0.0e+00	nan	0.0e+00	0.0e+00	0.0e+00	0.0e+00	nan	n

```
y_rep[72]              1.2e-02  2.0e-03  1.1e-01   0.0e+00   0.0e+00  0.0e+00  3.0e+03  8.1e-
y_rep[73]              6.3e-01  1.5e-02  8.2e-01   0.0e+00   0.0e+00  2.0e+00  3.1e+03  8.3e-
y_rep[74]              5.7e-02  4.6e-03  2.5e-01   0.0e+00   0.0e+00  1.0e+00  3.1e+03  8.3e-
y_rep[75]              1.3e-03  6.6e-04  3.6e-02   0.0e+00   0.0e+00  0.0e+00  3.0e+03  8.1e-
y_rep[76]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[77]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[78]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[79]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[80]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[81]              1.0e-03      nan  3.2e-02   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[82]              1.7e-02  2.3e-03  1.3e-01   0.0e+00   0.0e+00  0.0e+00  3.0e+03  8.0e-
y_rep[83]              3.1e+00  3.9e-02  2.0e+00   0.0e+00   3.0e+00  7.0e+00  2.6e+03  6.9e-
y_rep[84]              2.2e-02  2.9e-03  1.5e-01   0.0e+00   0.0e+00  0.0e+00  2.8e+03  7.5e-
y_rep[85]              1.3e-03      nan  3.6e-02   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[86]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[87]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[88]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[89]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[90]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[91]              1.0e-03  5.8e-04  3.2e-02   0.0e+00   0.0e+00  0.0e+00  3.0e+03  8.1e-
y_rep[92]              1.9e-02  2.6e-03  1.4e-01   0.0e+00   0.0e+00  0.0e+00  2.9e+03  7.8e-
y_rep[93]              5.9e+00  5.6e-02  3.0e+00   2.0e+00   6.0e+00  1.1e+01  2.9e+03  7.7e-
y_rep[94]              2.1e-02  2.6e-03  1.5e-01   0.0e+00   0.0e+00  0.0e+00  3.1e+03  8.5e-
y_rep[95]              3.3e-04      nan  1.8e-02   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[96]              3.3e-04      nan  1.8e-02   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[97]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[98]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[99]              0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n
y_rep[100]             0.0e+00      nan  0.0e+00   0.0e+00   0.0e+00  0.0e+00      nan      n

Samples were drawn using hmc with nuts.
For each parameter, N_Eff is a crude measure of effective sample size,
and R_hat is the potential scale reduction factor on split chains (at
convergence, R_hat=1).
```

```
#loo_results <- fit$loo(variables="lp__", cores=4)
#TODO: UHOH, why nan and what dis mean.
#print(loo_results)
```

We next move on to incorporating spatial correlation. To do so, we fit a Log Gaussian Cox Process (LGCP). Where before we specified the intensity as

$$\lambda(s) = \theta(s)b(s) = \exp[\alpha + \beta' x(s) + \gamma + \delta' z(s)]$$

we now add a Gaussian process

$$\lambda(s) = \theta(s)b(s) = \exp[\alpha + \beta' x(s) + \gamma + \delta' z(s) + w(s)]$$

so that the expected value for quadrat A will be the integral over the quadrat:

$$\Delta(A) = \int_A \lambda(s)ds = \int_A \exp[\alpha + \beta' x(s) + \gamma + \delta' z(s) + w(s)]ds$$

We make the same approximations as before based on our limited covariate resolution.

```
model_string <- log_gaussian_cox_process
write(model_string, model_path)
# Here we need to use the distance matrix
data = list(N = sites, X = grid_points$aux_x, Z = grid_points$aux_z, y = sim_data$Y[data_r
model <- cmdstan_model(model_path)
lgcp_fit <- model$sample(data=data, seed=13, chains=3, iter_sampling=10000, iter_warmup=10
```

```
lgcp_fit$summary(variables=c('alpha', 'beta', 'gamma', 'delta'))
```

```
posterior <- lgcp_fit$draws()
color_scheme_set("mix-blue-pink")
p_trace <- mcmc_trace(posterior,  pars = c("alpha", "beta", "gamma", "delta"),
                      facet_args = list(nrow = 2, labeller = label_parsed))
print(p_trace + facet_text(size = 15))

plot_title <- ggtitle(paste("Posterior distributions, with means and 90% interval"))
p_post <- mcmc_areas(posterior,  prob = 0.9, point_est="mean", regex_pars = c("alpha", "be
print(p_post)

plot_title <- ggtitle(paste("Posterior distributions, with means and 90% interval"))
p_post <- mcmc_areas(posterior,  prob = 0.9, point_est="mean", regex_pars = c("gamma", "de
print(p_post)

mcmc_intervals(posterior, pars=c('alpha', 'beta', 'gamma', 'delta'))
#mcmc_hist(fit$draws(), pars = c('alpha', 'beta', 'gamma', 'delta'))
plot_title <- ggtitle("Parameter posterior sample correlation")
p_post <- mcmc_pairs(posterior, pars=c('alpha', 'beta', 'gamma', 'delta'))
print(p_post)
mcmc_scatter(posterior, pars=c('alpha','gamma'))
```

```
generated_vars <- c('lambda_bias', 'lambda_rep', 'b_rep', 'y_rep')
lambda_thinned <- lgcp_fit$summary(variables=generated_vars[1])$mean
lambda_rep <- lgcp_fit$summary(variables=generated_vars[2])$mean
bias_rep <- lgcp_fit$summary(variables=generated_vars[3])$mean
y_ppd <- lgcp_fit$summary(variables=generated_vars[4])$mean
ppd_df <- data.frame(x=grid_points$x, y=grid_points$y, y_rep=y_ppd, lt=lambda_thinned, l=l
head(ppd_df)
```

We can then check the generated data:

```
p <- ggplot(ppd_df, aes(x, y, fill=y_rep)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("PPD counts per site")
print(p)

p <- ggplot(ppd_df, aes(x, y, fill=lt)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Thinned intensity per site")
print(p)

p <- ggplot(ppd_df, aes(x, y, fill=l)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Intensity per site")
print(p)
p <- ggplot(ppd_df, aes(x, y, fill=b)) +
    geom_tile() +
    scale_fill_viridis(discrete=FALSE) +
    ggtitle("Bias per site")
print(p)
```